Changes in Risk Factor Disclosures and the Variance Risk Premium

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Abstract: This paper examines how changes in risk disclosures affect uncertainty about risk. We measure changes in risk disclosures using the addition and removal of individual risk factors to firms' 10-K filings, identified via textual analysis of the risk factors section. Our market outcome is the variance risk premium (VRP), which captures the market's pricing of uncertainty about firm risk. Following recent theoretical predictions, we predict and empirically document that newly disclosed signals of risk-factor exposure—reflected in added and removed individual risk factors—decrease the uncertainty surrounding firm risk, as proxied via the VRP. We further confirm that individual risk factors offer incremental insights compared to alternative textual risk measures. Collectively, our findings suggest that textually evaluating individual risk factors reveals information about the uncertainty regarding firm risk.

Key Terms: risk factors, disclosure, textual analysis, variance risk premium

Data Sources: SEC filings, and the Compustat, CRSP, and OptionMetrics databases

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I. INTRODUCTION

This paper tests predictions from recent research (e.g., Heinle and Smith 2017) that firm-provided signals about risk factors lead to lower factor-exposure uncertainty, and thus lower uncertainty about firm risk. In particular, we examine whether changes in the composition of individual risk factors, identified via the application of textual analysis algorithms to the risk factor section of 10-K filings, reflect updated beliefs about firms' risk exposure, measured via the so-called variance risk premium, or VRP.[1]

To measure changes in firms' risk disclosures, we use the risk factors section of the 10-K filing, which offers a strong setting for our empirical tests for several reasons. First, the US Securities and Exchange Commission (SEC) mandated, effective 2005, that publicly-listed firms provide narrative discussion of all material risks the firm faces within the 10-K filing. Thus, we assume that the risk factors section of the 10-K represents a comprehensive source of material risks to the firm, noting that other channels (such as 8-K filings, MD&A sections, and earnings announcements) are more likely subject to selective inclusion of particular risks. Second, as a firm-supplied disclosure, the risk factors section benefits from managers' private information that can help investors identify and characterize the risk that the firm faces to an extent that they are unlikely able to replicate on their own. Finally, the SEC mandate requires each disclosed risk factor be labeled with a descriptive heading: we exploit this requirement using a textual analysis algorithm to identify individual risk factors—and changes in those risk factors—disclosed by the firm.

[1] As explained throughout the paper, the VRP is the difference between the cost of purchasing a contract to hedge future (realized) returns volatility and the subsequent realized volatility (Carr and Wu 2009). That is, it represents the compensation traders require to bear volatility risk.

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Prior academic research (e.g., Heinle and Smith 2017) suggests that signals provided by firms regarding their exposure to impending risks should reduce the *uncertainty* surrounding those risks. In particular, revealed signals about a risk factor provide inputs into market participants' assessment of the uncertainty about the variability of a firm's future cash flows, and facilitate investors' ability to more precisely estimate the effects of these risks on the firm, its performance, and the related stock price effects. Thus, risk-factor disclosures should reduce uncertainty about the second moment of cash flows and lead to more precise estimates of the distribution of future realized outcomes (i.e., realized cash flows). Accordingly, we hypothesize that firms providing greater changes in risk factors will exhibit lower uncertainty about their risk.

Testing this prediction requires two key proxies: one for a risk factor disclosure; and one for uncertainty about risk. As our proxy for changes in risk factor disclosures, we use the year-over-year *additions* and *removals* of individual risk factors within 10-K filings. We use textual analysis techniques to identify the risk caption headings provided in each firm's 10-K filing, and then apply string similarity algorithms to compare the text of the individual risk factor captions. This process allows us to isolate individual risk factors that have been added and those that have been removed relative to the previous year. We assume that additions and removals of risk factors provide unique information about management beliefs regarding changes in the exposure to various risks a firm faces. We then partition each year of observations into those characterized as having a high number of risk factor changes (proxied via those firms in the top quintile of total added and removed individual risk factors) and those having a low number of changes in individual risk factor disclosures (proxied via those firms in the bottom quintile).

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To proxy for investors' beliefs about uncertainty regarding risk, we use the VRP, which represents an estimable value directly related to the uncertainty about risk.[2] Prior empirical research (Carr and Wu 2009) documents that the cost of purchasing contracts to hedge future realized stock return volatility generally exceeds subsequent realized volatility, suggesting traders are willing to pay for a higher level of anticipated volatility than the actual volatility that occurs. This difference—sometimes denoted the "spread," or more formally, the VRP—represents the compensation traders require to bear volatility risk. Investors should only be willing to pay a premium to trade volatility if there is uncertainty about volatility itself; prior empirical research provides evidence consistent with this theoretical argument (Carr and Wu 2009). Accordingly, we use the VRP to proxy for market participants' uncertainty regarding the risks faced by the firms.

Using a broad cross-sectional sample spanning 2006-2019, empirical tests reveal a lower VRP for firms exhibiting higher additions and removals of individual risk factor disclosures. This finding is consistent with investors using changes in the disclosure of individual risk factors in firm's 10-K reports to update their beliefs about the uncertainty around the risks that firms face. We note two key research design choices. First, our tests employ recent matching procedures to minimize potential biases resulting from unobserved heterogeneity. Thus, after identifying firms having high versus low changes in individual risk factor disclosures, we match the high and low firms on a large set of control variables. This matching mitigates heterogeneity issues by minimizing differences in observable characteristics when comparing firms across risk factor disclosure groups. Second, our analyses directly control for word-based textual disclosure of both

[2] Throughout the study, we use the terms "uncertainty about risk" or "uncertainty about uncertainty" to align with the analytical settings of Heinle and Smith (2017) and Heinle, Smith, and Verrecchia (2018). In the option and asset pricing literatures, uncertainty about risk is sometimes called the "volatility of volatility," as it represents the uncertainty around future volatility. Theory suggests that for the VRP to exist, volatility itself must have volatility; accordingly, VRP represents a direct measure of the uncertainty about risk, which we wish to examine.

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the 10-K and the risk factors section used in prior research (e.g., Kravet and Muslu 2013; Campbell, Chen, Dhaliwal, Lu, and Steele 2014). This choice allows linking our findings to the addition and removal of individual risk factors, rather than to firms' disclosure policy or traditional measures of changes in risk-related words.

We confirm that our primary findings are robust to the following six sets of sensitivity analyses. First, results are consistent with alternative methods to derive the VRP, including expected (versus realized) variance measures and GARCH-based (versus non-GARCH-based) estimates. Second, results also are unchanged to using alternative thresholds to characterize high versus low changes in individual risk factor disclosures: terciles and medians (as opposed to quintiles in the primary analyses), suggesting that the effects we document are present throughout the range of available observations, not only in the tails of the distribution. Third, results are consistent to alternative matching procedures for the samples designated as high and low changes in individual risk factor disclosures, including propensity score matching as well as matching with replacement (versus without replacement in primary analyses). Fourth, our findings are robust to controlling for a variety of alternative text-based measures used in prior research (including Loughran and McDonald 2011, Kravet and Muslu 2013, and Campbell, Chen, Dhaliwal, Lu, and Steele 2014). We further document that these latter textual measures fail to exhibit a significant association with the VRP, suggesting our proposed individual risk factors proxy captures a unique disclosure attribute relative to these other measures. Fifth, results are unchanged to controlling for additional proxies for firms' economics. Finally, we employ placebo tests excluding our treatment variable and including the change in the number of words within the risk factor section. We fail to find significance for the latter variable, suggesting that the relation we document appears unique to changes in individually disclosed risk factors.

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We also conduct several cross-sectional tests motivated by Heinle and Smith (2017). First, we assess three firm characteristics. We predict and confirm incrementally stronger effects of risk factor additions and removals on the VRP for larger firms versus smaller firms, bellwether versus non-bellwether firms, and early-filing versus late-filing firms. Second, we assess three characteristics of the risk factors themselves. We separately assess risk factor additions and removals, documenting that both are negatively associated with the VRP. We also document incremental effects for those risk factors exhibiting the largest changes in materiality, proxied by exploiting our textual algorithms to identify the risk factors with the largest "jumps:" that is, those moving the most places within the risk factors section between consecutive years. We also confirm that the text associated with changed risk factors is associated with the VRP, not the text associated with "stale" (i.e., unchanged) risk factors. Finally, we conduct short-window analyses, providing evidence of more negative three-day abnormal VRP surrounding the 10-K filing for firms having the most added and removed risk factors, consistent with our primary longer-window analyses.

Our findings offer three central contributions. We provide the first direct empirical tests of the predictions of recent analytical research (e.g., Heinle and Smith 2017). This research argues that disclosing risk signals should reduce market participants' perceptions of the uncertainty about the firm's risk. Our findings are consistent with these predictions, using a market-based proxy (i.e., the VRP) that reflects uncertainty about such risks. Related, our work builds on Smith and So (2022), which measures the existence and timing of information about risk, by providing evidence of the source of that information. Second and related, we provide novel evidence about the decision-relevance of disclosed risk signals. In particular, we advance prior empirical papers, which document that word-based measures of risk disclosures are associated with higher perceived and realized riskiness (Kravet and Muslu 2013; Campbell et al. 2014) by demonstrating

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incremental information to identifying individual risk factors. Our findings show that company-sourced changes in disclosed risk factors reduce the *ex-ante* uncertainty surrounding future business outcomes. Third, we present a textual approach to identify and measure *individual* risk factors and several of their attributes. Our findings suggest that individual risk factors allow for a more complete characterization of the information underlying risk disclosures, with seemingly incremental and unique effects relative to more commonly-applied textual measures using word counts, and non-text measures such as volatility of income.

Section II provides the background and hypothesis development. Section III presents the research design. Section IV discusses the textual analysis methodology and descriptive statistics. Section V reports the primary empirical results, as well as validation and sensitivity analyses. Section VI presents cross-sectional and additional analyses, and Section VII concludes.

II. BACKGROUND AND HYPOTHESIS DEVELOPMENT

The Securities and Exchange Commission (SEC) dictates the reporting and disclosure requirements for US publicly-traded firms. In 2005, the SEC mandated that all firms provide a structured discussion of their risks within a dedicated section of their annual 10-K filings: *Item 1A, Risk Factors* (Regulation S-K, Item 503(c); SEC 2005). The requirements include listing and commenting on any material risk factors the firm faces that can impact future performance. Such factors reflect a wide range of risk categories, including those related to product development, customers, competitors, suppliers, and the legal and regulatory environments. Note that the SEC guidance specifies that the provided disclosures include risks specific to the registrant and exclude "generic" risks that could apply to any registrant (Kuntz, Ehrenberg, and Travis 2017).

Several other features of these risk factor disclosures warrant highlighting. First, the risk

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factors section represents a mandated SEC disclosure. While firms have discretion in identifying and characterizing "material" risk factors, the mandate mitigates endogenous reporting choices that can occur in other disclosure settings (e.g., other corporate narratives such as MD&A sections or CSR reports). Restated, the requirement to discuss *any* material risks incentivizes full disclosure, as failure to do so with subsequent *ex post* revelation can lead to significant costs such as litigation and/or negative compensation adjustments to senior management. Second, the SEC requires each individual risk factor be organized under a relevant sub-caption. This caption typically is a lead sentence summarizing the nature of the risk; in most filings, these captions are highlighted in some fashion (such as through bold-face, italics, or listed numerically). This latter feature facilitates our ability to identify and exploit the disclosure of individual risk factors (as described below).

Prior research documents that the textual content of the risk factors section has increased substantially over time (Dyer, Lang, and Stice-Lawrence 2017). This research generally characterizes risk-related textual information on an aggregate level, such as through total word or sentence counts within the risk factors section. Using such approaches, several studies assess the informativeness of aggregated risk disclosures. Campbell et al. (2014) finds that textual measures of risk disclosures, based on word counts, are related to lower information asymmetries (proxied using bid-ask spreads) and correlate with future systematic and idiosyncratic components of risk (proxied using CAPM Beta and stock return volatility). Kravet and Muslu (2013) further documents that changes in the number of risk-related words are positively associated with proxies of increased risk perception (including returns volatility, trading volumes, and the volatility of analyst forecast dispersions). Chiu, Guan and Kim (2018) extends these findings to credit markets, finding mandated SEC risk disclosures decreases uncertainty about companies' credit quality, as

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measured through credit default swaps spread and volatility. Brown et al. (2018) documents that SEC risk monitoring—evidenced through the regulators' comment letters—induces the monitored firm's peers to change their risk disclosures, consistent with reducing the likelihood of subsequent regulatory scrutiny. Dai, Landsman, and Peng (2022) reports that risk factor disclosures, measured through word counts and topic modeling, are positively associated with private debt pricing. Cohen, Malloy, and Nguyen (2020) finds that changes in 10-K risk factor disclosures, measured through string similarity algorithms such as cosine similarity, are a prominent driver of future stock returns; however, equity investors fail to fully price 10-K risk information when released, presumably due to the reports' length and complexity. Finally, Smith and So (2022) documents substantial risk information surrounding the release of 10-K filings, but also highlights the weak association between measures of risk disclosures based on word counts and risk information.[3]

We build on this literature by using textual analysis to identify *individual* risk factors within the risk factors section to assess investor uncertainty about uncertainty. In particular, using the mandated captions to discern changes in individual risk factors over time, we identify those factors that are static (i.e., unchanged from one year to the next), added, or removed. This provides granular changes in the disclosed risks of the firm, which we then relate to market measures. Critically, this also enables separate analysis of whether the over-time increase in aggregate risk disclosure or specific signals about risk factor exposure influence investor uncertainty surrounding firm risk. We assume that managers disaggregate their firms' risk narratives into individual risk factors to facilitate users' ability to identify and understand differential risks affecting the firm. Thus, we expect that the identification of individual risk factors conveys incremental information

[3] Other papers examine the role of accounting disclosures and option-implied risk measures, including Rogers, Skinner, and Van Buskirk (2009), Barth and So (2014), Billings and Jennings (2015), and Neururer, Papadakis, and Riedl (2016, 2020).

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beyond aggregate, word-based, text measures.[4] Accordingly, our research directly relates to prior work on the textual attributes of corporate narrative disclosures and their effects on financial and economic outcomes (Li 2008; Loughran and McDonald 2016). Our paper also builds on recent research characterizing the relevance of risk disclosures by focusing on individual risk factors as the unit of analysis. This includes Gaulin (2017), which argues that added risk factors help to predict firm's future adverse operating outcomes (such as earnings declines or losses), and Chin, Liu, and Moffitt (2018), which documents that highly ranked factors are positively correlated with higher credit risk (as measured using S&P ratings and credit spreads).

Prior theoretical research argues that signals about a firm's exposure to risks enable users of this information to adjust expectations of the variance of cash flows. In particular, Heinle and Smith (2017) argues that disclosures about firm risk allow investors to better estimate the distribution of future cash flows and hence the distribution of future stock prices. The paper suggests that higher moments of stock price outcomes can be utilized to characterize the nature of the information conveyed by risk factor disclosures, predicting that greater risk factor disclosure leads to lowered perceptions of the uncertainty about a firm's risk.

Prior literature (e.g., Carr and Wu 2009) uses the variance risk premium (VRP) to proxy for the market's perceived uncertainty about the risks a firm faces.[5] This reflects previous findings

[4] We note that newly promulgated regulation on risk factor disclosures (Release no. 33-10825; 34-89670; File No. S7-11-19, see also: sec.gov/news/press-release/2020-192) appears consistent with our expectation that individual risk factors convey incremental information beyond word counts. Specifically, the regulation requires (i) a summary (for especially lengthy disclosures), which we interpret as an attempt to synthesize and give prominence to risk factor headers, and (ii) a categorization; that is, an attempt to hierarchically group risk factors.

[5] We focus on the second moment of the return distribution, and not higher moments such as skewness or kurtosis, for several reasons. First, the prior research motivating our study (Smith and Heinle 2017; Heinle et al. 2018) models risk disclosures as information about the second moment. Second, the predicted associations are less clear with moments greater than two, and the empirical estimation of such moments is also more challenging (as it requires additional data and distributional assumptions). Related, we do not focus on testing empirical associations with respect to the first moment (i.e., the cost of capital), as Heinle et al. (2018) argues that the relation between the risk disclosure and such measures can be ambiguous.

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that option implied volatility generally exceeds subsequent realized volatility, a difference leading to a "spread" between option-implied and realized volatility and denoted the VRP. In asset pricing theory, the VRP reflects the premium investors are willing to pay to hedge against shocks to risk in future stock prices: the more uncertainty around this risk, the higher the VRP.

We assume that the addition or removal of individual risk factors provides signals that better enable market participants to characterize expectations of the uncertainty of a firm's risk outcomes. Restated, we use changes in risk factors to capture investor Bayesian updating of priors regarding risk. In this framing, the addition of a risk factor to Item 1A in the 10-K represents information allowing investors to update beliefs about risk, which they either previously were unaware of or did not perceive as material (as it was not previously included in Item 1A). Thus, we assume investors update priors by increasing their weighting on risk factors that are added, as the firm now indicates such factors to be material, relative to before the release of the 10-K. Similarly, the removal of a risk factor from Item 1A represents information enabling investors to update their beliefs about risk, which they may have previously viewed as material but is no longer viewed as such per management (as it is now excluded from Item 1A). Thus, we assume investors update their priors by decreasing their weighting on the removed risk factor, since it previously had been labeled as a material risk. As such, we predict that both additions and removals of a risk factor lead to updated risk perceptions, and thus less uncertainty regarding the material risks that the firm faces.[6] Accordingly, we predict that additions and removals of risk factors lead to lower uncertainty about risk (i.e., lower VRP), expressed in the following hypothesis:

[6] Since investors' expectations about individual risk factors are largely unobservable, we acknowledge that the level of risk factors can inform about uncertainty surrounding risk. However, we also note that the total number of risk factors is determined in large part by unchanged risk factors; these do not reflect evident changes in the management's beliefs regarding a firm's risk exposure. Therefore, we consider the level of risk factors to be less likely to foster a form of Bayesian updating of investors' priors relative to the change in risk factors.

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H1 Enhanced signals of individual risk factor exposure (proxied by additions and removals of individual risk factors to the risk factor section) are negatively associated with investor uncertainty of firm risk (proxied by the variance risk premium).

We note several sources of tension. While prior research provides evidence of risk disclosures correlating with market outcomes (Kravet and Muslu 2013; Campbell et al. 2014), we focus on the variance risk premium. VRP represents a unique variable that is non-redundant to traditional risk measures such as stock return volatility (Carr and Wu 2009), and provides an empirical proxy for the pricing of the theoretical construct of uncertainty about uncertainty. Thus, it is unclear that prior documented evidence of risk factor disclosures relation with various market risk outcome variables can be extrapolated to our VRP proxy. In addition, prior research questions whether firms simply satisfy the risk disclosure mandate through long and boilerplate discussions (IRRC 2016; SEC 2016). For example, Hope, Hu, and Lu (2016) argues that the informativeness of risk disclosures is directly related to their specificity and documents more significant market reactions to more specific disclosed risk factors. Beatty, Cheng, and Zhang (2019) challenges the informativeness of these disclosures, documenting a decrease in the relevance of risk discussions following the 2008 financial crisis, as well as lowered associations between risk narratives and future measures of realized operating risk (Altman's Z-Score). The latter paper suggests that this decline in informativeness may reflect companies' heightened focus on litigation concerns. Consistent with this latter claim, Cazier, McMullin, and Treu (2019) suggests that long and boilerplate risk discussions are effective as a shielding mechanism within judicial contexts. Thus, it is unclear from the extant research whether changes to risk disclosures are associated with uncertainty about risk.

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III. RESEARCH DESIGN

To test how changes in risk factor disclosures affect market perceptions of the firm's factor-exposure uncertainty, we use the following research design:

$$
\begin{aligned}
VRP_{jt} = \; & \alpha_1 + \beta_1 \#RF\ Add/Rem_{jt} \\
& + \beta_2 MA\ Ret\ Volatility_{jt} + \beta_3 Beta_{jt} + \beta_4 CFO\ Volatility_{jt} + \beta_5 Size_{jt} \\
& + \beta_6 ROA_{jt} + \beta_7 Loss_{jt} + \beta_8 BM_{jt} + \beta_9 Leverage_{jt} \\
& + \beta_{10} Analyst\ Follow_{jt} + \beta_{11} Inst\ Investors_{jt} + \beta_{12} Text\ Length\ 10K\ wo\ RF_{jt} \\
& + \text{Year-Month F.E.} + \text{Industry F.E.} + \varepsilon_{jt} \qquad (1)
\end{aligned}
$$

The dependent variable is the variance risk premium (*VRP*), measured as the natural log of the one-month ahead option-implied variance less the natural log of a GARCH(1,1)-based one-month ahead realized variance. Prior empirical research (Carr and Wu 2009) documents that the cost of purchasing contracts to hedge future realized stock return volatility is generally higher relative to subsequent realized volatility. This "spread" between the cost of the "variance contract" and future realized variance is the VRP, representing the compensation traders require to bear volatility risk. For a fixed level of expected future volatility, higher option implied volatility (and thus higher VRP) indicates a higher perceived uncertainty about future volatility.[7]

[7] The VRP can be estimated comparing option-implied variance to either *realized* or *expected* variance. We calculate implied variance at the end of each month for 30-day ahead expiration options from OptionMetrics volatility surface files applying the so-called "model-free" approach using both "at-" and "out-of-the-money" strikes (e.g., Barth and So 2014; Callen and Lyle 2020). Realized variance can be calculated using a model-based (e.g., GARCH-based model as in Bakshi and Kapadia 2003) or non-model-based (e.g., from daily squared log-returns) approach. Expected variance is usually estimated through a model-based approach. We use a GARCH model as it (i) facilitates both *forecasting* and *inferring* realized variance, and (ii) offers useful statistical features in the presence of infrequent volatility shocks. In particular, GARCH models' use of large series of historical data allow extreme volatility shocks to converge and become negligible. We estimate the GARCH monthly using one year (approximately 252 trading days) of historical data. Results are unchanged to using both *realized* and *expected* variance, and both *GARCH* and *non-GARCH* estimations.

We also note that Carr and Wu (2009) presents VRP as realized less option-implied variance. We swap the terms in this subtraction (e.g., as in Bollerslev et al. 2009) to facilitate interpretation; accordingly, in our analyses lower (higher) values of *VRP* indicate lower (higher) levels of expected uncertainty about future firm risk.

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The treatment variable is the change in disclosed risk factors, *#RF Add/Rem*, measured as the absolute number of added and removed risk factors from firm *j*'s 10-K "Item 1A" from filing year *t-1* to *t*. Our sample selection includes firm-years exhibiting the highest (top quintile, representing our treatment group) and lowest (bottom quintile, representing our benchmark group) changes in added plus removed risk factors. Thus, this variable measures new signals of the firm's riskiness to financial statement users, proxied via the addition and removal of individual risk factors. Our focus on groups of firms ranked on risk disclosure quintiles parallels the "portfolio-group" approach common in empirical asset pricing literature (e.g., Fama and French 1993). Ranking equates to non-parametric cross-sectional estimation (Cochrane 2011), accommodates non-linear relations with assets returns, and fosters more interpretable results. We predict that firms with the highest addition and removal of individual risk factors provide investors with the most signals—thus leading to decreased perceived uncertainty about a firm's risk, and a negative predicted coefficient on *#RF Add/Rem*. As a second proxy, we examine *#RF Add/Rem Scaled*: the absolute added plus removed risk factors, scaled by the total number of risk factors in year *t-1*. Scaling mitigates potential misclassification due to heterogeneity from different total disclosed risk factors levels. Our empirical approach is similar to that of Francis, LaFond, Olsson, and Schipper (2005), which regresses future market outcomes onto metrics of information quality.

We include the following control variables (Sridharan 2015). First, we include four proxies for historical risk. *MA Ret Volatility* is the standard deviation of firm *j*'s monthly value-weighted market-adjusted stock returns, measured over the preceding 12 months. We also include *Beta* as a proxy for firm *j*'s systematic risk; it is the firm's beta derived from a one-factor market model estimated over the year preceding the 10-K filing. We include *CFO Volatility* to proxy for the firm's operating risk, measured as the standard deviation of firm *j*'s annual operating cash flows

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(each scaled by beginning total assets) over the preceding five years. Finally, we include *Size*, the natural logarithm of firm *j*'s market capitalization at the beginning of the filing year *t*.

Second, we include variables to control for differences across firms. We include *ROA*, firm *j*'s net income for year *t* divided by beginning total assets. We also include *Loss*, an indicator variable equal to 1 if firm *j* reports negative net income for year *t*, and 0 otherwise. Both control for the effects of performance on perceived riskiness. We include *BM*, firm *j*'s book-to-market ratio measured at the beginning of the filing year *t*, to control for the firm's growth opportunities. We include *Leverage*, firm *j*'s short-term plus long-term liabilities divided by total assets at the beginning of filing year *t*, to control for financial constraints. We then include two measures of the firm's information environment: *Analyst Follow* (the natural logarithm of the maximum number of earnings estimates for firm *j* in filing year *t*), and *Inst Investors* (the number of shares held by institutional investors in firm *j* for year *t*, scaled by total shares outstanding).

Third, we include *Text Length 10K wo RF*, the natural log of the total words in firm *j*'s 10-K for year *t* less the "Item 1A" Risk Factors total words for the same filing. This variable captures the propensity towards disclosure in the 10-K filing; also, longer filings may reflect higher business and/or linguistic complexity. Additional specifications control for *ΔRF Text Length* (measured as the change in the total number of words within "Item 1A") to confirm that added and removed risk factors capture a dimension of disclosures *distinct from changes in the number of risk words*.[8] All variables are defined in Appendix A. We include date and industry fixed effects, and cluster standard errors by firm and date (e.g., Campbell et al. 2014; Beatty et al. 2019).

[8] Note that we expect *ex ante* that the number of risk-related words will be a noisier measure of uncertainty regarding risk relative to our primary variable based on individual risk factors. Later robustness tests confirm this expectation, revealing a negative but insignificant association between changes in the number of risk words and the VRP.

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Finally, to rule out confounding differences between firms displaying a "high" and a "low" number of added and removed individual risk factors, we match the firms as follows. First, we perform an exact match on industry, trading month, and the loss binary variable to ensure comparable groups of firms belonging to the same industry, time period, and profit versus loss characterization. Second, we match firms on the remaining Equation (1) control variables, capturing (i) drivers of observed risk (*MA Ret Volatility*; *Beta*; *CFO Volatility;* and *Size*), (ii) fundamentals (*ROA*; *BM*; *Leverage*; *Analyst Follow*; and *Inst Investors*), and (iii) characteristics of the textual filing (*Text Length 10K wo RF*). Our primary tests use a one-to-one "Nearest-Neighbor" matching, based on "Mahalanobis Distance," with no replacement.[9] We then entropy-balance (Hainmueller 2012; Zhao and Percival 2016) the first and second moments (i.e., mean and variance) of the continuous variables' distributions to level remaining differences between the two firm groups and ensure results are not driven by heterogeneity in other characteristics across the high and low risk factor groupings.[10]

IV. DATA AND DESCRIPTIVE STATISTICS

Data and Textual Analysis Methodology

We use three primary data sources. Textual data is extracted from 10-K filings obtained from the *SEC EDGAR* database. Firms' financial fundamentals are from *WRDS Compustat*

[9] Untabulated results are also robust to propensity score matching and matching with replacement.

[10] We opt for a conservative stepwise approach to maximize similarity across covariates between "high" and "low" risk factor disclosers by (i) matching observations through "Nearest-Neighbor", (ii) adjusting their weights through entropy-balancing, and (iii) estimating weighted OLS regressions on the matched-pair samples (Ruhose, Thomsen, and Weilage 2019). Note that results are similar to (a) not applying entropy-balancing and only using "Nearest-Neighbor" matching, or (b) directly applying entropy-balancing with no prior matching (following Hainmueller 2012). Thus, our presented (sequential) use of both "Nearest-Neighbor" matching and entropy-balancing provides a conservative estimation. Of note, our primary estimation leads to a non-negligible number of unmatched observations due to (i) exact matching on industry, trading month, and loss indicator, and (ii) a small balancing tolerance (i.e., only observations with a balancing loss below 1e-6 are matched). For completeness and robustness, we report results (unchanged) using a full-panel unweighted regression within the Online Appendix (Table OA3).

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(annual files) and *CRSP* (monthly files). Variance risk premia (VRP) are estimated using *OptionMetrics*'s volatility surface files and CRSP daily files. We use OptionMetrics data to estimate option-implied variance, and the CRSP files to estimate expected and realized variance.

First, we identify and download 10-K reports filed 2006–2019 with valid matches in financial databases (i.e., valid CIK, GVKEY, and PERMNO numbers).[11] We scrape each document to extract a company's identifiers, the filing date, the reporting period, "Item 1A" (i.e., the "Risk Factors" section), and individual risk-factor disclosures within "Item 1A." To minimize measurement error, we discard 10-K filings for which "Item 1A" and/or individual risk factors are difficult to extract; we also exclude a filing if the lagged risk factor disclosures have not been identified or the relevant financial items are missing.[12] Our sample comprises 33,797 annual filings. Each firm-year is then matched with monthly VRP data, leading to 180,449 monthly observations with available risk factor disclosures, financial fundamentals, and VRP (see Table 1). Our final sample (50,330 matched-pair observations) matches firms with a *high* (fifth quintile) to a *low* (first quintile) number of added and removed risk factor disclosures.[13]

We quantify the total number of added and removed risk factors in a firm's 10-K report by developing textual analysis procedures that: (i) identify "Item 1A" within 10-K filings; (ii) extract individual risk factors within "Item 1A"; and (iii) classify each individual risk factor as *added* (i.e.,

[11] As the SEC requires the disclosure of risk factors within annual reports in 2005, we begin our sample with 2006.

[12] Difficult-to-extract "Item 1A" sections result from missing or non-standard starting/ending markers used to parse the textual reports. Difficult-to-extract disclosed risk factors result from non-standard formatting and layout within "Item 1A" of 10-K reports. Thus, the main textual treatment variable (the number of added and removed risk factors for a given 10-K report) requires the existence and identification of two consecutive (i) annual filings, (ii) "Item 1A" sections, and (iii) risk-factor disclosures (i.e., individual risk factors).

[13] We use monthly VRP and annual risk disclosures in our main tests for two reasons. First, using monthly dependent variables and yearly independent variables is common in the empirical asset-pricing literature (e.g., Fama and French, 1992, 1993, 2020; see also Chattopadhyay, Lyle and Wang 2022 and the references within). Second, longer-dated options data (e.g., yearly maturities) is more sparsely populated in the OptionMetrics database, and the associated options tend to be less liquid than those having shorter expirations. Our maintained assumption is that changes in perceived uncertainty about uncertainty are priced regardless of horizon. Results are unchanged to using average monthly VRP over a filing year as our market measure: see Online Appendix Table OA 4.

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if missing in the prior filing period) or *removed* (i.e., if missing in the current filing year). We first download the "HTML" version of all available 10-K reports from *SEC EDGAR*; we exclude 10-K Amendments, 10-KSB, and 10-KSB Amendments. We then scrape each document to identify a set of starting signals (e.g., "1A", "RISK FACTORS") and ending signals (e.g., "1B", "UNRESOLVED"); this enables us to delimit and extract the "Risk Factors" section.[14]

To identify and extract individual risk factors within "Item 1A", we exploit the textual layout and formatting preserved in the "HTML" file. Risk factors typically are introduced through a boldface and/or italicized header, followed by a plain-text paragraph(s) containing an in-depth discussion of the individual risk factor. Accordingly, we classify individual risk factors as portions of the text within "Item 1A" that are both (i) introduced by a boldface and/or italicized header, and (ii) followed by a plain-text paragraph.[15]

A novelty of our study is linking individual *additions* and *removals* of risk factors (our proxy for change in risk factor disclosures) to VRP (our proxy for uncertainty about risk). To identify *added* and *removed* risk factors, we exploit flexible string-similarity algorithms to assess whether a risk factor is repeated within two consecutive 10-K reports (see Appendix B).[16] We classify a risk factor as "added" if it was not disclosed in the prior year but is disclosed in the

[14] We also develop tailored conditional statements to handle cases of multiple starting and/or ending signals. Whenever the "Item 1A" cannot be identified with sufficient reliability, we exclude the document from the sample; this leads to the exclusion of 17% of the total downloaded 10-K reports. In addition, we confirm the reliability of the parsing strategy by manually checking the extracted "Item 1A" for 50 distinct companies in different years.

[15] In some cases, boldface and/or italicized headers introduce the "type" or "class" of risk factors (e.g., "Risks Related to Our Business and Industry") rather than an individual risk factor. Our algorithm excludes such cases, as we require a plain-text paragraph to follow the header for classification as an individual risk factor.

[16] Specifically, we combine several *Python* algorithms based on *Levenshtein Distance* to assess string similarity: (i) regardless of the position of the tokens found within two textual sequences (so-called "token sort ratio" approach); (ii) despite differences in two strings that are solely driven by repeated tokens (so-called "token set ratio" approach); and (iii) even if minor other differences are present between the two sequences we compare (e.g., punctuation, special characters). Each algorithm outputs a similarity score between 0% and 100%; we take the average of all similarity scores and identify a "match" for scores above 85%. We manually inspect 100 "matched" risk factors and 100 "unmatched" risk factors for different firm-years and find two false positives and one false negative case, resulting in about 99% accuracy.

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present year; conversely, a "removed" risk factor was disclosed in the prior year but not in the current year. Figure 1 provides descriptive data on added and removed risk factors across disclosure quintiles. In Panel A, firm-years in the top (bottom) quintile added or removed an average of 15.8 (less than 1) risk factors over the full sample period; per Panel B, this represents a 50% (2%) change relative to the mean of 32 risk factors disclosed across the sample period.[17]

Descriptive Statistics

Table 2 presents the descriptive statistics for the primary sample of matched-pair observations, with the full sample in Column (1), "high-disclosing" subsample of Quintile 5 in Column (2), and "low-disclosing" subsample of Quintile 1 in Column (3). Consistent with needed options data, the sample tilts towards larger firms (mean market capitalization of $6.1 billion). The mean *ROA* is 1%; 24% of firms exhibit losses; the mean book-to-market ratio is 0.51; and the sample firms exhibit moderate leverage. Column (4) presents differences across the "high-disclosing" and "low-disclosing" subsamples after matching and entropy balancing. We observe no differences in means or standard deviations across any control variables, which mitigates potential confounds from other characteristics that may correlate with our treatment variable. As expected, the dependent variable (*VRP*) has a significantly lower mean (i.e., lower uncertainty

[17] We note that measuring individually added and removed risk factors through textual analysis underlies two key trade-offs. First, this measurement is more complex to implement than traditional word counts, but it is also less noisy to identify "news" (i.e., information affecting investors' expectations) within 10-K risk disclosures. For example, a company may exhibit a limited change in the number of risk-related words (from filing year "*t*-1" to filing year "*t*"), while substantially changing its risk factor disclosures, such as if the same number of risk factors (with a similar number of words describing them) are added and removed (i.e., two individual risk factors are added and two are removed), or the position of existing factors changes substantially (thus management is signaling that the weighting of the underlying risks should be revised). This motivates our development and use of *changes* in risk factor disclosures based on individual risk factors. Second, our measurement requires a string-similarity threshold to identify risk factors as unchanged versus added/removed. A stricter threshold (i.e., requiring high similarity, such as 95% versus the used 85%) will reduce Type I (increase Type II) errors. Conversely, a more moderate threshold (i.e., requiring relatively lower similarity, such as 75% versus the used 85%) will increase Type I (reduce Type II) errors. Finally, a strength of our measurement is that it enables a machine-based approach to identifying comprehensive individual risk factors for a broad cross-section of firms.

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about risk) for the "high disclosing" (0.41) relative to the "low-disclosing" (0.46) subsample. We report unweighted sample statistics (i.e., after matching but before entropy balancing) within the Online Appendix, Table OA 2. Matching alone reduces the average covariates' difference by roughly 70%. However, the remaining differences appear statistically significant, supporting our choice to additionally entropy balance the covariates' distribution.

Table 3 presents the correlations. Both *#RF Add/Rem* and *#RF Add/Rem Scaled* exhibit significantly negative correlations with *VRP*, consistent with lower perceived uncertainty about firm risk for firms with the highest number of added or removed risk factors.

V. EMPIRICAL RESULTS

Primary Analyses

Table 4 presents the primary empirical results. Column (1) contains results including only the treatment variable of *#RF Add/Rem* and fixed effects, revealing the expected significantly negative coefficient (–0.22; *t*-stat = –2.62). Column (2) repeats the results including the four control variables to capture the firm's risk environment (*MA Ret Volatility*, *Beta*, *CFO Volatility*, and *Size*). The coefficient on *#RF Add/Rem* remains significantly negative (–0.20; *t*-stat = –2.67). Column (3) presents results including the additional firm-level control variables (*ROA*, *Loss*, *BM*, *Leverage*, *Analyst Follow*, *Inst Investors*, and *Text Length 10K wo RF*). Again, the coefficient on *#RF Add/Rem* is significantly negative (–0.23; *t*-stat = –3.04).

Column (4) presents results including one additional control variable: *ΔRF Text Length*, the change in the number of words within the 10-K "Item 1A". We include this variable to discern if perceived uncertainty about risk reflects changes in individual risk factors (as captured by *#RF Add/Rem*) and/or changes in aggregate, word-based, text disclosure in the risk factors section (as

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captured by *ΔRF Text Length*). The coefficient on *#RF Add/Rem* remains significantly negative (–0.22; *t*-stat = –2.96); the coefficient significance and signs on the other control variables are unchanged. Of note, the coefficient on *ΔRF Text Length* is negative but insignificant (–0.02; *t*-stat = –0.72). Combined with the insignificant coefficient on *Text Length 10K wo RF*, this suggests neither 10-K disclosure propensity, nor traditionally-evaluated changes in risk words, drive perceived uncertainty about risk. More importantly, the results also suggest that the changes in individual risk factors capture information incremental to changes in total risk words. Finally, Column (5) presents results replacing the treatment variable with the scaled proxy of *#RF Add/Rem Scaled*. Results are unchanged: the coefficient on *#RF Add/Rem Scaled* is significantly negative (–6.19; *t*-stat = –2.90), and the significance and coefficients on the controls variables are unchanged.

Combined, the results suggest that market participants reassess the firm's riskiness conditional on observing the disclosure of newly added and removed individual risk factors. These results are consistent with predictions of Heinle and Smith (2017) that the disclosure of additional risk signals (proxied via added and removed risk factors) reduces perceptions of forward-looking uncertainty about risk (proxied via the VRP), and support *H1*.

Validation and Sensitivity Analyses

Validation Analyses

One premise of our study is that additions and removals of individual risk factors capture new information about uncertainty surrounding risk. Therefore, we expect our measure to associate with notions of risk information alternative to VRP. Accordingly, we follow Smith and So (2022) and compute their proposed *RiskInfo* measure, and then estimate portfolio tests (see their Table 7), sorting companies into five yearly quintile portfolios of *RiskInfo*. In Table 5, consistent with Smith

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and So (2022), we fail to find evidence of a significant difference (t = 1.00) in the average portfolio change of risk words within "Item 1A" of 10-K reports for companies characterized by high (i.e., fifth quintile) versus low (i.e., first quintile) *RiskInfo*. In contrast, we find a significantly positive difference (t = 3.81) in additions and removals of individual risk factors for high versus low *RiskInfo* companies. In economic terms, companies with high *RiskInfo* exhibit, on average, a 10% larger change in individual risk factors relative to firms with low *RiskInfo*. Overall, our evidence suggests additions and removals of individual risk factors relate to an alternative notion of risk information developed by Smith and So (2022).[18]

Sensitivity Analyses: Alternative Dependent Variables

Our primary analyses use GARCH-based inferred realized variance to estimate the *realized* VRP as the dependent variable. A potential concern is that our results are sensitive to this measurement of stock return variance. Accordingly, we examine two alternative VRP estimations. First, we use expected (i.e., forecasted) VRP (*VRP_Expected*) based on the natural logarithm of Option Implied Variance minus the one-month ahead variance forecast. Using expected VRP relies entirely on *ex-ante* information, mitigating potential confounds of using realized (*ex post*) information (as done in our primary analyses). Second, we use a realized VRP based on a non-GARCH estimation: *VRP_Non-Garch*, the natural logarithm of Option Implied Variance minus that of one-month ahead (non-GARCH) realized variance using the sum of daily squared log-returns.[19] As it also is common to use "model free" variance estimation of realized volatility, we

[18] We conduct several additional tests. First, we use *RiskInfo* as the dependent variable in a short-window regression analysis: while we find generally significantly positive associations between additions and removals of risk factors and *RiskInfo* surrounding a 10-K, this result is sensitive to the inclusion of the full set of controls used in our main regressions. Second, we use alternative measurements of risk words (as described in subsequent sensitivity analyses) and find results similar to those presented in Table 5.

[19] The correlations across the three dependent variables (i.e., *VRP*, *VRP_Expected*, and *VRP_Non-Garch*) range from 0.43 to 0.77, suggesting similar but not entirely overlapping measurements of uncertainty regarding risk.

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conduct this estimation to ensure our results are not driven by our choice of the volatility model. Table 6 Panel A presents the results. In Columns (1) and (2) using *VRP_Expected*, we find significantly negative coefficients on *#RF Add/Rem* (–0.21; *t*-stat = –2.49) as well as *#RF Add/Rem Scaled* (–5.09; *t*-stat = –2.22). In Columns (3) and (4) using *VRP_Non-Garch*, we again find significantly negative coefficients on both *#RF Add/Rem* (–0.12; *t*-stat = –2.17) and *#RF Add/Rem Scaled* (–4.04; *t*-stat = –2.24). Untabulated coefficient signs and significance levels on the control variables are similar to those reported in Table 4. Overall, the results appear robust to other dependent variable measures.

Sensitivity Analyses: Alternative Sample Partitions

Next, we investigate how widely the documented associations occur within the distribution of available observations. The primary analyses define the change in individual risk factor disclosure using the top and bottom quintiles of the sample distribution. We now assess two alternative derivations, both of which expand the sample of included observations. First, we use terciles, with high (low) firm groupings defined as firms in the top (bottom) tercile of added plus removed risk factors (*N* = 81,142 matched-pair observations). Second, we use medians, with high (low) firm groupings defined as firms above (below) the median of added plus removed risk factors (*N* = 131,634 matched-pair observations). Table 6 Panel B presents the results. In Columns (1) and (2) using tercile partitions, we find significantly negative coefficients on both *#RF Add/Rem* and *#RF Add/Rem Scaled*; similar results obtain in Columns (3) and (4) using median partitions. Overall, the results appear robust to widening the included sample of firms.[20]

[20] We note the coefficient magnitudes on the treatment variables decrease as the sample is broadened, suggesting the strongest effects are in the tails of total added and removed individual risk factors. Results also are consistent to (i) propensity score matching (see Hirano, Imbens, and Ridder 2003) with no replacement, and (ii) estimating one-

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Sensitivity Analyses: Comparisons with Other Text-Based Risk Measures

Following prior textual analysis methodologies, we compare the informativeness of our risk disclosure measure (*#RF Add/Rem*) against eight alternative proxies capturing changes in risk disclosures to ensure our measure is not subsumed by them. First, we consider *ΔRF Text Length*, the change in risk factor text length, to proxy for the change in total risk factor disclosure within "Item 1A" (Dyer et al. 2017). Next, we calculate four proxies based on risk-specific dictionary approaches (i.e., we use a subset of risk-specific words to compute the measures): *ΔRF Text Length – LM*, the change in uncertainty-related words identified using the Loughran and McDonald (2011) dictionary approach (used in Smith and So 2022); *ΔRF Text Length – CEA*, the change in risk-related words identified through the Campbell et al. (2014) dictionary (used in Beatty et al. 2019); *ΔRF Text Length – CEA (id)*, the change in words related to idiosyncratic risk identified using the Campbell et al. (2014) dictionary of idiosyncratic risk words; and *ΔRF Text Length – KM*, the change in risk-related words identified using the Kravet and Muslu (2013) dictionary. Finally, we derive three proxies for changes in 10-K risk factor disclosures using *cosine similarity* algorithms (as used in Brown and Tucker 2010 and Cohen et al. 2020): *ΔCosine RF Text*, one minus the cosine similarity of "Item 1A" within 10-K filings between consecutive years; *ΔCosine RF Text (Heads)*, one minus the cosine similarity of (only) risk factor headers text within "Item 1A" of 10-K filings; and *ΔCosine RF Text (Pars)*, one minus the cosine similarity of (only) risk factor paragraphs text within "Item 1A" of 10-K filings. We apply a cosine similarity protocol to stand-alone headers text to ascertain whether a relatively simpler measure of change in risk factor disclosure subsumes the specific identification of additions and removals of individual risk factors. We then re-estimate our primary regressions, alternatively controlling for each of the above eight text-based risk

to-one nearest neighbor based on "Mahalanobis Distance", but now with replacement to provide potentially stronger matches at the cost of potentially overstated significance due to increased correlations across observations.

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disclosure measures, to ensure our findings are incremental to these other proxies. Table 7 reveals that our treatment variable (*#RF Add/Rem*) remains significantly negative across all eight specifications, and that none of the other text-based risk measures is significant. Combined, the results suggest our individual risk factors offer incremental insights for investors' pricing of uncertainty about risk, relative to previously-examined text measures of risk.

Sensitivity Analyses: Alternative Explanations

Finally, we assess three potential alternative explanations to our findings. First, we re-estimate our primary analyses by also including the contemporaneous (*Earn Surprise (t)*) and one-year ahead (*Earn Surprise (t+1)*) earnings surprises to control for any economic effects not already subsumed by our primary control variables, as Sridharan (2015) finds that earnings information is a key determinant of option returns and stock volatility. Table 8 reveals our results remain consistent to including only *Earn Surprise (t)* in Column (1), *Earn Surprise (t+1)* in Column (2), both earnings surprises in Column (3), and with a match on both the earnings surprises in Column (4). Second, the primary treatment variable focuses on measuring *individual* risk-factor signals, rather than the total number of risk words. We next conduct untabulated placebo tests using the eight alternative disclosure proxies based on changes in the total number of risk-related words (described above) as the treatment variables, while excluding changes in individual risk factors (i.e., *#RF Add/Rem*). This assesses whether changes in risk factor text length may spuriously drive the observed relationships. Results confirm that the coefficients on 7 of 8 of the alternative disclosure proxies are insignificant, with one (i.e., that of *ΔRF Text Length – LM*) being weakly

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significant.[21] This evidence suggests that the individual risk factors within "Item 1A" appear to capture a distinct association with uncertainty about risk, and that the signed change in the number of risk words (examined by prior research) does not align well with the notion of information reflected in VRP. Finally, we conduct an analysis using an unweighted full panel regression, alternatively adding industry and firm fixed effects: results remain consistent with our primary analyses (see the Online Appendix Table OA 3).

VI. CROSS-SECTIONAL AND ADDITIONAL ANALYSES

Cross-Sectional Analyses

We now examine if the negative association between added/removed risk factors and the VRP varies with predictable cross-sectional attributes. First, we examine three firm characteristics. Heinle and Smith (2017) primarily concerns information on systematic variance; thus, larger firms are expected to exhibit stronger effects relative to smaller firms, due to the former having more systematic risk. Accordingly, we partition our sample into larger and smaller firms (proxied via above versus below annual median market capitalization), and re-estimate our regressions separately on the two samples. Table 9 Panel A presents the results. Column (1) using the larger firm sample reveals a significantly negative coefficient on *#RF Add/Rem* (–0.29; *t*-stat = –3.09); Column (2) using the smaller firm sample reveals a negative but insignificant coefficient (–0.10; *t*-stat = –0.92), with the larger firm sample coefficient being more negative (*p*-value = 0.02).

Second, we identify firms more likely to be characterized as bellwether using the technique of Bonsall et al. (2013), which suggests that bellwether firms provide informational spillovers due

[21] Results are unchanged to (i) running the same analyses excluding control variables, and (ii) defining the treatment variable to parallel our primary findings (i.e., equal to 1 for observations in the top quintile, and 0 if in the bottom quintile, of each alternative disclosure proxy distribution).

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to the timing of their filings and/or roles within the economy. Column (3) using the bellwether firm sample reveals a significantly negative coefficient on *#RF Add/Rem* (–0.39; *t*-stat = –5.34); Column (4) using the non-bellwether firms reveals an insignificant coefficient (–0.15; *t*-stat = –1.36); and as expected the coefficient is more negative for the bellwether sample (*p*-value = 0.02).

Third, we conduct analyses partitioning firms into early versus late 10-K filers. We expect a stronger association for earlier filers, which are more likely to reveal new information in their risk factor disclosures. We define early (late) filing firms as those below (above) the median days between the fiscal year-end and the 10-K filing. Column (5) using the sample of early filing firms reveals a significantly negative coefficient on *#RF Add/Rem* (–0.36; *t*-stat = –3.56); Column (6) using the late filing firms, shows that coefficient to be insignificant (–0.11; *t*-stat = –1.06); and as predicted the coefficient is more negative for the early filing sample (*p*-value < 0.00).

Next, we examine cross-sectional characteristics of the risk factors themselves. The primary analyses combine additions and removals of risk factors; we now decompose these separately into the number *added* (*#RF Added*) and *removed* (*#RF Removed*). Table 9 Panel B presents the results, with Column (1) revealing a significantly negative coefficient on *#RF Added* (–0.31; *t*-stat = –2.84) and Column (2) a significantly negative coefficient on *#RF Removed* (–0.31; *t*-stat = –2.42); the coefficients are not statistically different (untabulated). We interpret this evidence as consistent with both additions and removals representing a form of Bayesian updating to investors regarding uncertainty about risk. That is, the addition of a risk factor can inform investors of a risk they may previously have been unaware of, or did not perceive as material (since it was previously not disclosed in the 10-K within "Item 1A"). Similarly, the removal of a risk factor can update investor priors about a risk they previously viewed as material, but is now no longer viewed as such per management, leading to a decreased weighting on the removed risk

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factor (since it previously had been labeled as a material risk). Related, our results offer novel empirical insights into uncertainty surrounding risk as opposed to risk. Beatty et al. (2019) documents that added (discontinued) risk disclosures increase (decrease) risk perceptions, with both measured through word counts. In contrast, our above findings demonstrate that *both* additions and removals of risk signals reduce perceptions of risk uncertainty, suggesting risk and uncertainty about risk are distinct constructs (Carr and Wu 2009).

Second, we examine a further dimension of change for individual risk factors. Specifically, we derive a proxy assessing the change in materiality of existing (rather than added or removed) risk factors (e.g., Gaulin 2017). We rely on the 10-K preparation guidelines recommending that individual risk factors be ordered in terms of economic significance; best practice suggests many firms do so voluntarily to minimize litigation risks (Gelfond, Wechsler, and Cohen 2017).[22] Accordingly, we use the changes in the positioning within the risk factors section to proxy for changes in the materiality of the risk signals (see Online Appendix, Table OA 1, for an example). We create proxies capturing the percentage of risk factors exhibiting movement within the risk factors section by five or more places (*%RF Jump 5+*) or by three or four places (*%RF Jump 3-4*).[23] We expect that higher movement suggests greater changes in materiality and, thus stronger signals informing investors about risk uncertainty. Column (3) uses the larger movements, with the coefficient on *%RF Jump 5+* being significantly negative (–0.08; *t*-stat = –2.07). Column (4) uses the smaller movements, with an insignificant coefficient on *%RF Jump 3-4* (–0.05; *t*-stat = –

[22] The SEC approach to risk factor disclosures is consistent with other contexts such as Form 20-F for non-US firms.
[23] We do not use a proxy for movement by one or two places, as the latter are more likely subject to mechanical position changes that can occur when a factor is added or removed.

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0.83). We note that both results are incremental to the primary treatment variable of *#RF Add/Rem Scaled*, which remains significantly negative across both specifications.[24]

Third, we partition the total risk factor text (denoted *Text Length (RF)*) into the portion attributable to changes in risk factors (i.e., additions, removals, or material changes in their positioning; denoted *Text Length (ΔRF)*) versus that for risk factors not exhibiting material changes (denoted *Text Length ("Stale RF")*). Column (5) reveals that total risk factor text length exhibits a negative coefficient (–0.04; *t*-stat = –2.86). Of note, we continue to control for *#RF Add/Rem Scaled*, which exhibits a significantly negative association with VRP (–5.49; *t*-stat = –2.67). Column (6) then shows a significantly negative coefficient on *Text Length (ΔRF)* (–0.01; *t*-stat = –3.99), and insignificant coefficient on *Text Length ("Stale RF")* (–0.01; *t*-stat = –0.77). These results confirm that the reduction in uncertainty about uncertainty derives primarily from text relating to changing risk factors versus pre-existing and presumably unchanged risk factors.

Combined, the cross-sectional analyses suggest that the negative effect of added and removed risk factors on the VRP is accentuated for larger firms, bellwether firms, and those filing their 10-K reports earlier. In addition, the effect occurs for both added and removed risk factors, is accentuated for those risk factors exhibiting the highest movement within the risk factors section, and is concentrated on text relating to changing (as opposed to "stale") risk factors.[25]

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[24] As an additional risk factor attribute, we consider using the contents of the risk factors to identify those more likely to reflect uncertainty about risk, as opposed to downside risk. Detailed review of several risk factors reveals challenges to such a characterization. For example, consider a pharmaceutical firm adding a risk factor indicating that a particular drug is coming off patent (i.e., the so-called "patent cliff"), which ostensibly appears to be a "left-tail" type risk. However, closer inspection of these disclosures reveals that firms typically also discuss other information that can be interpreted as "right-tail" risk, including: (i) plans to renew the existing drug patent, (ii) initiatives to revamp the existing patent into a related drug to extend its economic life, and (iii) other related therapies in development that will replace/improve upon the existing drug. This example illustrates the challenges to systematically characterizing a given risk factor as primarily relating to downside versus two-tailed risk.

[25] Table 9 results are similar if the total number of risk factors is used as a control or to scale the treatment variable.

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Additional Analyses

We conduct three additional analyses. First, we estimate a short-window test. While prior research documents market participants' delayed reaction to risk factor disclosures (Cohen et al. 2020), the use of long sample windows in our primary analyses makes it difficult to attribute a causal effect of risk factor additions and removals to reduced uncertainty about firm risk. Prior work provides evidence of short-window informational effects in the option markets regarding earnings announcements (Savor and Wilson 2016; Hann, Kim, Zheng 2019). We build on these findings to improve identification, and conduct a short-window analysis similar to Barth and So (2014) to test for reductions in uncertainty about risk surrounding the 10-K release. Specifically, we develop an abnormal variance risk premium as follows. We measure the event VRP in two alternative ways: (a) as the three-day [–1, +1], and (b) two-day [0, +1] VRP surrounding the 10-K release (where day "0" corresponds to the 10-K filing date). We define the non-event period as the three trading weeks preceding the event, and measure the non-event VRP either as: (i) a random three-day and two-day VRP drawn from the non-event window (following Barth and So 2014); or (ii) the median three-day and two-day VRP across the non-event window. We then re-estimate our regression, now using *Abn_VRP* as the dependent variable—alternatively defined as the event VRP less each of the two above non-event VRP, to assess whether the abnormal VRP exhibits the predicted negative association with added/removed risk factors. Table 10 reveals significantly negative coefficients on *#RF Add/Rem* using both the random and median non-event VRP as the benchmark, consistent across both the three-day (Panel A) and two-day (Panel B) event windows. The findings also are similar using the unconditional regressions in Columns (1) and (4), including the Table 4 controls of Columns (2) and (5), and additionally including the earning surprise in Columns (3) and (6). To gauge economic significance, we sort firms in yearly quintiles of

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abnormal VRP and find that, on average, companies in the first quintile disclose about 20% more additions and removals of risk factors relative to firms in the fifth quintile. Overall, our results are consistent with incrementally reduced VRP in the three-day and two-day windows surrounding the 10-K release, for those 10-K reports having more added or removed risk factors.[26, 27]

Second, we estimate three additional specifications. In the first two, we hold constant the VRP preceding the 10-K release. In particular, we: (i) include as controls *both* the VRP of the preceding twelve months, and that of the month before the filing date; and (ii) express the dependent variable in changes (the difference between current month VRP and the prior twelve months VRP). The intent is to assess if changes in risk factor disclosures are associated with post-filing VRP *conditioning* on pre-filing VRP. In the third, we average monthly VRP over the filing year to mitigate concerns associated with the mismatched frequency of the dependent and control variables. Results (unchanged) are in the Online Appendix (Table OA 4).

Third, we conduct the following untabulated analysis assessing the association between changes in risk factor disclosures and future stock returns. Heinle and Smith (2017) suggests that greater risk factor disclosures should raise expected stock returns upon the disclosure release (assuming a constant level of risk), and thus lower future expected returns. However, Heinle et al. (2018) also discusses the ambiguity of risk disclosures on expected returns: for example, the addition of a risk factor can increase perceived risk (e.g., due to the market now pricing a previously unknown risk) or decrease perceived risk (e.g., due to the market now viewing the risk as lower relative to the *ex ante* perception). Untabulated results are directionally consistent with

[26] Untabulated analyses are similar alternatively using the mean three/two-day VRP across the non-event window.

[27] We note some caveats to conducting a short-window test in our setting. As documented by Arif et. al. (2019), earnings and 10-K reports are often released within the same three/two-day windows. While we include both the current and forward (i.e., 1-year ahead) earnings surprises to control for cash-flow news, we note an inherent difficulty in separating earnings-based information from information that is independent within the 10-K filing. In robustness tests, we exclude filing events that fall within 3 days of an earnings announcement; this reduces our sample by 20%. Results are similar and statistically significant, though generally weaker.

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the reduction in uncertainty around the second moment, with limited evidence that future returns are lower for firms having more added and removed risk factors.[28]

VII. CONCLUSION

This paper examines whether risk disclosures are related to uncertainty about firm risk. Our proxy for changes in the disclosed risk signals uses the text within "Item 1A" ("Risk Factors") of firms' 10-K filings, enabling the identification of firm-level additions and removals of individual risk factors. Using the variance risk premium (VRP) as a theoretically-motivated measure of investor perceptions of the uncertainty of firm risks, we document that companies with the highest changes to individual risk factor disclosures exhibit lower VRP relative to those having the lowest changes to individual risk factor disclosures. The findings are consistent with market participants using additions and removals of individual risk factors to form expectations of uncertainty about future risk outcomes. Our work offers new insights regarding the decision-relevance of risk factor disclosures: company-sourced risk signals not only map into future realized variance, but also reduce *ex ante* uncertainty surrounding future risks. Critically, we confirm that our proposed measure of individual risk factors is not subsumed by extant textual risk measures using word counts, suggesting that our measure captures a distinct dimension of risk relative to these other proxies. Future research may explore the attributes/contents of the individual risk factors, time-series trends in the reporting of individual risk factors, risk factors' mapping into future horizons and longer-dated options, and the consequences on other outputs by market participants such as institutional investors, creditors, and intermediaries like analysts.

[28] In untabulated analyses we also rank the VRP each month based on disclosure quintiles (consistent with the "portfolio-group" approach), and test the average monthly difference in realized VRP between "high" (i.e., 5th quintile) and "low" (i.e., 1st quintile) risk-factor disclosers, finding a significant difference of –0.038 (*t*-stat = –4.59). This evidence corroborates our main finding.

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APPENDIX A – Variable Descriptions

	Description	Source
Dependent Variable		
VRP	Natural logarithm of Option Implied Variance (estimated using the "model-free" approach while using Gatheral and Jacquier's (2014) SSVI to fit the volatility slice and extrapolate over strikes, while remaining arbitrage free) minus natural logarithm of GARCH(1,1)-based one-month ahead realized variance	OptionMetrics and CRSP
Treatment Variables		
#RF Add/Rem	Absolute number of added and removed risk factors (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings	EDGAR
#RF Add/Rem Scaled	Absolute number of added and removed risk factors (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings scaled by the total number of risk factors in year *t-1*	EDGAR
Control Variables		
MA Ret Volatility	12-month standard deviation of a firm's monthly market-adjusted stock returns, with market return proxied via the value-weighted index including dividend distributions	CRSP
Beta	CAPM beta, using monthly returns over the past 12 months	CRSP
CFO Volatility	5-year standard deviation of annual operating cash-flows divided by total assets	Compustat
Size	Natural logarithm of a firm's market capitalization	CRSP
ROA	Net income/total assets	Compustat
Loss	Indicator equal to 1 if net income is negative, and 0 otherwise	Compustat
BM	Book value per share / fiscal closing price	Compustat
Leverage	(Short-term liabilities + long term liabilities) / total assets	Compustat
Analyst Follow	Natural logarithm of the maximum number of estimates in a filing year divided by the natural logarithm of total assets; missing observations are replaced with a zero value	I/B/E/S
Inst Investors	Number of shares held by institutional investors each year divided by the total number of shares outstanding	Reuters 13-F
Text Length 10K wo RF	Natural logarithm of the total words in a 10-K minus the total words within "Item 1A"	EDGAR
ΔRF Text Length	Change in the total number of words within "Item 1A" from filing year *t-1* to *t*, divided by 1,000	EDGAR
Earn Surprise (t)	EPS in 10-K filing quarter *q* less the mean quarter *q* EPS forecast made prior to the earnings announcement and 10-K filing dates, scaled by stock price at the end of quarter *q*	I/B/E/S
Earn Surprise (t+1)	EPS in 10-K filing quarter *q+4* (i.e., next-year) less the mean quarter *q+4* EPS forecast made prior to the earnings announcement and 10-K filing dates, scaled by stock price at the end of quarter *q+4*	I/B/E/S
Variables for Validation, Sensitivity Analyses, and Cross-Sectional Tests		
RiskInfo	Smith and So (2022, page 392) absolute *RiskInfo*, based on 30-day standardized options and centered on 10-K filing dates	OptionMetrics and CRSP
VRP_Expected	Natural logarithm of Option Implied Variance minus that of Garch-based one-month ahead forecast of variance	OptionMetrics and CRSP

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VRP_Non-Garch	Natural log of Option Implied Variance less that of one-month ahead realized variance using sum of daily squared log-returns	OptionMetrics and CRSP
Abn_VRP _Random	Abnormal VRP computed as the non-event VRP (a random three/two-day—[*t-1*, *t+1*] and [*t*, *t+1*], where *t* is the filing date—VRP drawn from the non-event window) less the event VRP (the three/two-day VRP surrounding the 10-K release)	OptionMetrics and CRSP
Abn_VRP _Median	Abnormal VRP computed as the non-event VRP (the median three/two-day—[*t-1*, *t+1*] and [*t*, *t+1*] where *t* is the filing date—VRP in the non-event window) less the event VRP (the three/two-day VRP surrounding the 10-K release)	OptionMetrics and CRSP
#RF Added	Absolute number of added risk factors (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings	EDGAR
#RF Removed	Absolute number of removed risk factors (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings	EDGAR
%RF Jump 5+	Percentage of risk factors that "jump" (i.e., move within the risk-factor listing) by five or more positions (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings	EDGAR
%RF Jump 3-4	Percentage of risk factors that "jump" (i.e., move within the risk-factor listing) by three or four positions (from filing year *t-1* to *t*) within "Item 1A" of 10-K filings	EDGAR
ΔRF Text Length – LM	Change in the number of uncertainty-related words, using the Loughran and McDonald (2011) dictionary, in "Item 1A" from filing year *t-1* to *t*, divided by 1,000	EDGAR
ΔRF Text Length – CEA	Change in the number of risk-related words, using the Campbell et al. (2014) dictionary, in "Item 1A" from year *t-1* to *t*, divided by 1,000	EDGAR
ΔRF Text Length – CEA (id)	Change in the number of idiosyncratic risk words, using the Campbell et al. (2014) dictionary, in "Item 1A" from year *t-1* to *t*, divided by 1,000	EDGAR
ΔRF Text Length – KM	Change in the number of risk-related words, using the Kravet and Muslu (2013) dictionary, in "Item 1A" from year *t-1* to *t*, divided by 1,000	EDGAR
ΔCosine RF Text	One minus the cosine similarity of "Item 1A" between filing year *t-1* and *t*, using all words within the risk factor sections. All cosine similarity measures use a design matrix based on raw word frequencies with a TF-IDF adjustment of less weight on very frequent words (e.g., articles and prepositions such as "the", "a", "and")	EDGAR
ΔCosine RF Text (Heads)	One minus the cosine similarity of "Item 1A" between filing year *t-1* and *t*, using only the risk factor headers	EDGAR
ΔCosine RF Text (Pars)	One minus the cosine similarity of "Item 1A" between filing year *t-1* and *t*, using only the risk factor paragraphs	EDGAR
Text Length (RF)	The natural logarithm of the number of words within "Item 1A" (i.e., "Risk Factors") from filing year *t-1* to *t*	EDGAR
Text Length (ΔRF)	The natural logarithm of the number of words for changing risk factors within "Item 1A" (i.e., added, removed, or exhibit material positional changes from filing year *t-1* to *t*)	EDGAR
Text Length ("Stale"RF)	The natural logarithm of the number of words for unchanged risk factors within "Item 1A" (i.e., neither added nor removed and no material positional changes from filing year *t-1* to *t*)	EDGAR

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APPENDIX B – Example of Textually-Evaluated Changes in Risk Factor Disclosures

Chevron Corporation (CIK: 93410) – Individual Risk Factors Within 10-K "Item 1A"

	2009	2010	2011
(1)	*Chevron is exposed to the effects of changing commodity prices.*	*Chevron is exposed to the effects of changing commodity prices.*	*Chevron is exposed to the effects of changing commodity prices.*
(2)	*The scope of Chevron's business will decline if the company does not successfully develop resources.*	*The scope of Chevron's business will decline if the company does not successfully develop resources.*	*The scope of Chevron's business will decline if the company does not successfully develop resources.*
(3)	*The company's operations could be disrupted by natural or human factors.*	*The company's operations could be disrupted by natural or human factors.*	*The company's operations could be disrupted by natural or human factors.*
(4)	*Chevron's business subjects the company to liability risks.*	*Chevron's business subjects the company to liability risks from litigation or government action.*	**The company's operations have inherent risks and hazards that require significant and continuous oversight.**
(5)	*Political instability could harm Chevron's business.*	*Political instability could harm Chevron's business.*	*Chevron's business subjects the company to liability risks from litigation or government action.*
(6)	*Regulation of greenhouse gas emissions could increase Chevron's operational costs and reduce demand for Chevron's products.*	*Regulation of greenhouse gas emissions could increase Chevron's operational costs and reduce demand for Chevron's products.*	*Political instability could harm Chevron's business.*
(7)		**Changes in management's estimates and assumptions may have a material impact on the company's consolidated financial statements and financial or operations performance in any given period.**	*Regulation of greenhouse gas emissions could increase Chevron's operational costs and reduce demand for Chevron's products.*
(8)			*Changes in management's estimates and assumptions may have a material impact on the company's consolidated financial statements and financial or operations performance in any given period.*
Additions and Deletions (bolded items above):			
	0	**1**	**1**

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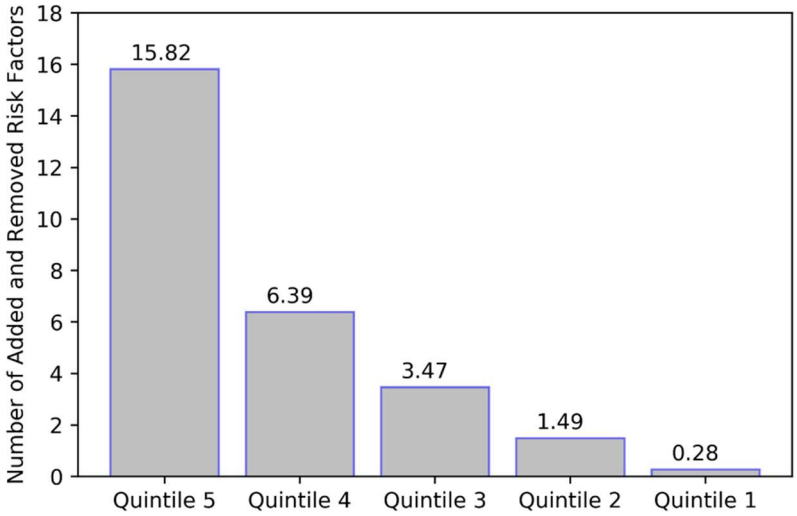
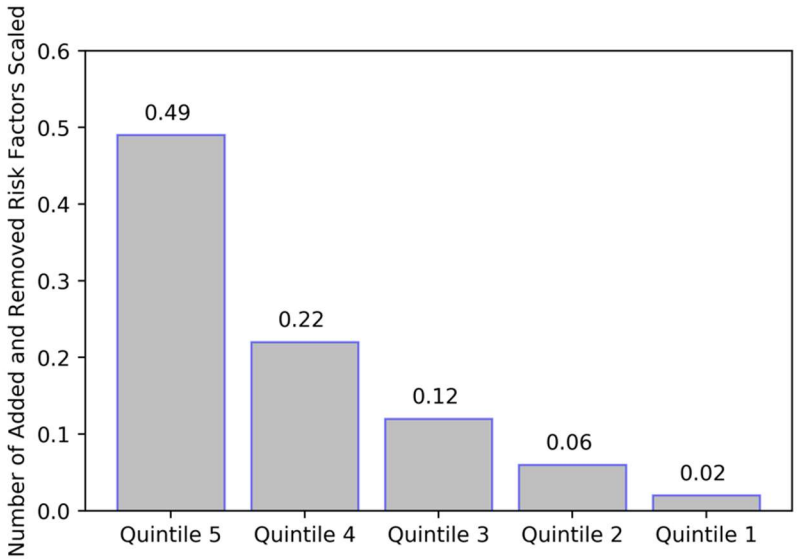
FIGURE 1

Panel A: Number of Added and Removed Risk Factors Within 10K Item 1A



Panel B: Scaled Number of Added and Removed Risk Factors Within 10K Item 1A



This figure presents the average number of added and removed risk factors (Panel A) and the average number of added and removed risk factors scaled by the lagged (i.e., in *t-1*) number of risk factors (Panel B) found within "Item 1A" (i.e., "Risk Factors") of SEC EDGAR 10-K reports, filed 2006–2019. The total number of firm-year observations is 49,740 (representing 54,773 available firm-year disclosures less 5,033 missing lagged disclosures). Five bars are displayed, one for each quintile of the cross-sectional (i.e., by filing year) distribution of scaled added and removed risk factors, assessed over the full sample period. The number of absolute and scaled added and removed risk factors are winsorized at the 1st and 99th percentiles of their cross-sectional distribution.

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TABLE 1
Sample Selection

	Number
EDGAR 10-K filings (2006-2019) by companies with valid CIK, GVKEY, and PERMNO	72,322
Less: 10-K filings with:	
- difficult-to-extract Item 1A (i.e., "Risk Factors")	(12,551)
- difficult-to-extract disclosed risk factors within Item 1A	(4,998)
Total EDGAR 10-K filings with available risk-factor disclosures	54,773
Less: observations with missing:	
- Compustat items	(8,220)
- CRSP items	(7,723)
- lagged disclosures to compute textual variables	(5,033)
Total firm-years with available risk-factor disclosures and financial data	33,797
Total firm-year-months with available risk-factor disclosures, and financial data	180,449
Primary sample of matched-pairs observations (by quintiles)	**50,330**
"High-disclosing" (i.e., top quintile) firm-year-months	25,165
"Low-disclosing" (i.e., bottom quintile) firm-year-months	25,165

This table presents the sample selection for the final sample of matched-pairs observations (i.e., firm-year-months) with available data, including (i) available risk disclosures from SEC EDGAR 10-K "Item 1A," (ii) financial fundamentals, and (iii) and variance risk premia ("VRP," obtained from OptionMetrics). "Difficult-to-extract Item 1A sections" result from missing or non-standard starting/ending markers used to parse the textual reports. "Difficult-to-extract disclosed risk-factors" result from non-standard formatting and layout within "Item 1A" of 10-K reports. Missing "lagged disclosures to compute textual variables" result from the absence of firm-specific prior period disclosed risk factors, which are necessary to compute the total number of added and removed risk-factor disclosures between consecutive filing years. Yearly SEC EDGAR textual disclosures are joined with monthly OptionMetrics VRP.

The primary matching strategy is a one-to-one "Nearest-Neighbor" approach (based on "Mahalanobis Distance") and estimated without replacement. The matched-pairs sample comprises 50,330 observations, and includes firms disclosing a *high* number (5th quintile of the yearly distribution) and *low* number (1st quintile of the yearly distribution) of added and removed risk factors within Item 1A ("Risk Factors") of 10-K filings.

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TABLE 2
Descriptive Statistics

	Full Matched Sample (N = 50,330)		"High-disclosing" Quintile 5 Sample (N = 25,165)		"Low-disclosing" Quintile 1 Sample (N = 25,165)		Difference "High" – "Low" (Post Entropy-Balancing)	
	(1)		(2)		(3)		(4)	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Dependent Variable								
VRP	0.43	0.65	0.41	0.66	0.46	0.65	–0.05***	0.01***
Treatment Variables								
#RF Add/Rem	7.83	8.71	15.10	6.75	0.56	0.57	14.54***	6.18***
#RF Add/Rem Scaled	0.26	0.29	0.48	0.25	0.03	0.04	0.45***	0.21***
Matched Control Variables								
MA Ret Volatility	0.11	0.07	0.11	0.01	0.11	0.01	0.00	0.00
Beta	1.06	0.89	1.06	0.89	1.06	0.89	0.00	0.00
CFO Volatility	0.05	0.06	0.05	0.06	0.05	0.06	0.00	0.00
Size	7.33	1.53	7.33	1.53	7.33	1.53	0.00	0.00
ROA	0.01	0.18	0.01	0.18	0.01	0.18	0.00	0.00
Loss	0.24	0.18	0.24	0.18	0.24	0.18	0.00	0.00
BM	0.51	0.39	0.51	0.39	0.51	0.39	0.00	0.00
Leverage	0.23	0.21	0.23	0.21	0.23	0.21	0.00	0.00
Analyst Follow	0.29	0.11	0.29	0.11	0.29	0.11	0.00	0.00
Inst Investors	0.83	0.25	0.83	0.25	0.83	0.25	0.00	0.00
Text Length 10K wo RF	10.82	0.47	10.82	0.47	10.82	0.47	0.00	0.00

This table presents descriptive statistics for the primary sample of 50,330 matched-pairs observations. The matching reflects firm-year-month observations chosen within the highest and lowest quintiles of added and removed risk factors within "Item 1A" (i.e., "Risk Factors") of 10-K reports filed 2006–2019. All variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles. "Dependent" ("Treatment") Variables reflect the primary dependent and treatment variables. "Matched Control Variables" include the covariates used to match "high" risk-factor to "low" risk-factor observations. The matching approach is one-to-one "Nearest-Neighbor" (based on "Mahalanobis Distance") and estimated without replacement. Exact matching is performed on year, month, industry membership, and the loss indicator; the mean and variance of the "Matched Control Variables" are entropy-balanced. Columns 1 (2) [3] present mean and standard deviations for the full matched sample (sample disclosing a *high* number of added and removed risk factors) [sample disclosing a *low* number of added and removed risk factors]; Column (4) presents tests of differences in mean and standard deviation across the two subsamples. *** indicates significance at <0.01 (two-tailed test).

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TABLE 3
Correlations

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1)	*#RF Add/Rem*	1													
(2)	*#RF Add/Rem Scaled*	**0.80**	1												
(3)	*VRP*	**–0.02**	**–0.03**	1											
(4)	*MA Ret Volatility*	**0.10**	**0.04**	**–0.06**	1										
(5)	*Beta*	**0.01**	**–0.01**	**–0.06**	**0.23**	1									
(6)	*CFO Volatility*	**0.02**	**–0.03**	**–0.02**	**0.29**	**0.09**	1								
(7)	*Size*	**–0.04**	**0.02**	**–0.16**	**–0.45**	**–0.06**	**–0.30**	1							
(8)	*ROA*	**–0.04**	**0.02**	**–0.03**	**–0.31**	**–0.06**	**–0.48**	**0.30**	1						
(9)	*Loss*	**0.06**	**–0.03**	0.00	**0.40**	**0.09**	**0.26**	**–0.37**	**–0.53**	1					
(10)	*BM*	**0.05**	**0.08**	**0.11**	**0.02**	**0.04**	**–0.13**	**–0.21**	**0.05**	**0.04**	1				
(11)	*Leverage*	**0.02**	**0.03**	**0.01**	**0.04**	**0.05**	**–0.03**	**0.11**	**–0.06**	**0.04**	**–0.14**	1			
(12)	*Analyst Follow*	**–0.06**	**–0.10**	**–0.14**	**–0.03**	**–0.02**	**0.01**	**0.15**	**–0.01**	**0.02**	**–0.22**	**–0.09**	1		
(13)	*Inst Investors*	**–0.02**	**0.02**	**–0.18**	**–0.07**	**0.03**	**–0.11**	**0.16**	**0.17**	**–0.12**	0.00	**0.06**	**0.31**	1	
(14)	*Text Length 10K wo RF*	**0.12**	**0.06**	0.00	**–0.07**	**0.02**	**–0.10**	**0.30**	**–0.02**	**0.01**	**0.10**	**0.21**	**–0.06**	0.00	1
(15)	*ΔRF Text Length*	**0.12**	**0.27**	**–0.02**	**–0.01**	**–0.02**	**–0.04**	**0.03**	**0.01**	**–0.01**	**0.03**	**0.01**	**–0.02**	**0.02**	**0.02**

This table reports Pearson correlations for the primary sample of 50,330 matched-pair observations; the matching reflects firm-year-month observations chosen within the highest and lowest quintiles of added and removed risk factors within "Item 1A" (i.e., "Risk Factors") of 10-K reports filed between 2006–2019. All variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles. Bold coefficients indicate significance at <0.01 (two-tailed tests).

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TABLE 4
Risk Factor Disclosures and the Variance Risk Premium

		Dependent Variable: *VRP*				
	H1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
#RF Add/Rem	**(–)**	**–0.22 *****	**–0.20 *****	**–0.23 *****	**–0.22 *****	
		(–2.62)	**(–2.67)**	**(–3.04)**	**(–2.96)**	
#RF Add/Rem Scaled	**(–)**					**–6.19 *****
						(–2.90)
MA Ret Volatility			–1.18 ***	–1.21 ***	–1.21 ***	–1.22 ***
			(–6.50)	(–7.02)	(–7.02)	(–7.03)
Beta			–0.02 ***	–0.02 ***	–0.02 ***	–0.02 ***
			(–2.87)	(–2.96)	(–2.96)	(–2.94)
CFO Volatility			–0.40 ***	–0.43 ***	–0.43 ***	–0.43 ***
			(–3.00)	(–3.20)	(–3.22)	(–3.19)
Size			–0.12 ***	–0.11 ***	–0.11 ***	–0.11 ***
			(–16.54)	(–13.60)	(–13.59)	(–13.48)
ROA				–0.08	–0.08	–0.08
				(–1.57)	(–1.58)	(–1.58)
Loss				–0.03	–0.03	–0.03
				(–1.28)	(–1.29)	(–1.31)
BM				0.01	0.01	0.01
				(0.47)	(0.46)	(0.47)
Leverage				0.01	0.01	0.01
				(0.31)	(0.32)	(0.31)
Analyst Follow				–0.53 ***	–0.53 ***	–0.54 ***
				(–5.46)	(–5.47)	(–5.50)
Inst Investors				–0.16 ***	–0.16 ***	–0.16 ***
				(–4.40)	(–4.40)	(–4.38)
Text Length 10K wo RF				–0.02	–0.02	–0.02
				(–0.91)	(–0.92)	(–1.06)
ΔRF Text Length					–0.02	–0.01
					(–0.72)	(–0.34)
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry	Date Industry
N		50,330	50,330	50,330	50,330	50,330
Adjusted R^2		0.16	0.22	0.23	0.23	0.23

This table presents OLS results examining the effect of risk factor disclosures on the variance risk premium. The sample includes matched-pair observations, reflecting firm-year-month observations chosen within the highest and lowest quintiles of added and removed risk factors within "Item 1A" (i.e., "Risk Factors") of 10-K reports filed 2006–2019 ("filing years"). The matching process is described in Table 2. The dependent variable is the realized variance risk premium (*VRP*). The treatment variables are *#RF Add/Rem* (*#RF Add/Rem Scaled*), defined as the absolute number of added and removed risk factors within a 10-K "Item 1A" (the number of added and removed risk factors scaled by the total number of lagged risk factors). The coefficients of the experimental variables are multiplied by 100. All other variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10, respectively (two-tailed tests).

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TABLE 5
Validation Analyses: Smith and So (2022) *RiskInfo* Portfolio Tests

Tested Variables	***High-Low RiskInfo Quintile Portfolios (Mean)***			***High-Low RiskInfo Quintile Portfolios (Median)***		
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
#RF Add/Rem	**0.80 *****			**0.86 *****		
	(3.81)			**(4.84)**		
#RF Add/Rem Scaled		**0.02 ***			**0.02 *****	
		(1.87)			**(2.96)**	
ΔRF Text Length			0.43			0.12
			(1.00)			(0.85)
Sample	Full Panel	Full Panel	Full Panel	Full Panel	Full Panel	Full Panel
Portfolio	Yearly Quintile	Yearly Quintile	Yearly Quintile	Yearly Quintile	Yearly Quintile	Yearly Quintile
Construction	Mean	Mean	Mean	Median	Median	Median

This table presents validation analyses comparing for statistical differences in means. These differences are tested for the following three variables: *#RF Add/Rem*, *#RF Add/Rem Scaled*, and *ΔRF Text Length*. The comparison involves the top (i.e., 5th) and bottom (i.e., 1st) yearly quintile portfolios sorted based on the Smith and So (2022) absolute *RiskInfo* measure, computed using 30-day standardized options and centered on 10-K filing dates. In Columns (1)-(3) the yearly portfolios are based on mean yearly values; in Columns (4)-(6) the yearly portfolios are based on median yearly values. Differences between high and low portfolios and associated *t*-statistics are reported in all columns; untabulated Wilcoxon-Mann-Whitney tests are similar. The reported analyses are based on the full unweighted panel sample. All variables are defined in Appendix A. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE 6
Sensitivity Analyses: Alternative Dependent Variables and Sample Partitions

Panel A: Alternative Dependent Variables

		Dependent Variable:			
		VRP_Expected		***VRP_Non-Garch***	
	H1	**(1)**	**(2)**	**(3)**	**(4)**
#RF Add/Rem	**(–)**	**–0.21 ****		**–0.12 ****	
		(–2.49)		**(–2.17)**	
#RF Add/Rem Scaled	**(–)**		**–5.09 ****		**–4.04 ****
			(–2.22)		**(–2.24)**
Controls		Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry
N		50,286	50,286	50,330	50,330
Adjusted R^2		0.20	0.20	0.26	0.26

Panel B: Alternative Sample Partitions

		Dependent Variable: ***VRP***			
		Terciles		**Median**	
	H1	**(1)**	**(2)**	**(3)**	**(4)**
#RF Add/Rem	**(–)**	**–0.22 *****		**–0.21 *****	
		(–3.10)		**(–3.08)**	
#RF Add/Rem Scaled	**(–)**		**–4.56 ****		**–3.31 ***
			(–2.39)		**(–1.84)**
Controls		Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry
N		81,142	81,142	131,634	131,634
Adjusted R^2		0.24	0.23	0.23	0.23

This table presents sensitivity OLS analyses examining the effect of risk factor disclosures on the variance risk premium. Panel A presents results using two alternative dependent variables: expected VRP based on Garch-based one-month-ahead forecasted variance (i.e., *VRP_Expected*) in Columns (1) and (2), and realized VRP based on one-month ahead (non-Garch) realized variance derived from the sum of daily squared returns (i.e., *VRP_Non-Garch*) in Columns (3) and (4). Panel B uses alternatively defined matched-pair samples of firms disclosing a high number versus low number of added and removed risk factors within "Item 1A" of 10-K reports filed 2006–2019 based on highest and lowest tercile in Columns (1) and (2), and above (below) the median in Columns (3) and (4). Across both panels, the treatment variables are *#RF Add/Rem* (*#RF Add/Rem Scaled*), defined as the absolute number of added and removed risk factors within a 10-K "Item 1A" (the number of added and removed risk factors scaled by the total number of lagged risk factors).

The coefficients of the experimental variables are multiplied by 100. All other variables are defined in Appendix A. Controls include the full set of control variables used in Table 4. The matching process is described in Table 2. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE 7
Sensitivity Analyses: Comparisons with Other Text-Based Risk Measures

	Dependent Variable: *VRP*							
Text-Based Measure:	***ΔRF Text Length***	***ΔRF Text Length – LM***	***ΔRF Text Length – CEA***	***ΔRF Text Length – CEA (id)***	***ΔRF Text Length – KM***	***ΔCosine RF Text***	***ΔCosine RF Text (Heads)***	***ΔCosine RF Text (Pars)***
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
#RF Add/Rem	**–0.22 *****	**–0.22 *****	**–0.22 *****	**–0.22 *****	**–0.22 *****	**–0.29 *****	**–0.33 *****	**–0.29 *****
	(–2.96)	**(–2.93)**	**(–2.96)**	**(–2.97)**	**(–2.94)**	**(–3.30)**	**(–3.13)**	**(–3.32)**
Text-Based Measure	–0.02	–0.11	–0.03	–0.04	–0.09	0.15	0.11	0.15
	(–0.72)	(–1.40)	(–0.99)	(–0.82)	(–1.23)	(1.40)	(1.48)	(1.44)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry
Observations	50,330	50,330	50,330	50,330	50,330	50,330	50,330	50,330
Adjusted R^2	0.23	0.23	0.23	0.23	0.23	0.24	0.24	0.24

This table presents results from sensitivity analyses examining the effect of added and removed risk factors within "Item 1A" of the 10-K on the variance risk premium, relative to alternative text-based risk measures. The dependent variable is the realized variance risk premium (*VRP*). The treatment variable is *#RF Add/Rem*, defined as the absolute number of added and removed risk factors within a 10-K "Item 1A." The coefficients of the treatment variable are multiplied by 100. The other text-based measures include the following: *ΔRF Text Length*, the year-on-year change in the total words within the risk factor section of the 10-K; *ΔRF Text Length – LM*, the change in the number of uncertainty-related words, using the Loughran and McDonald (2011) dictionary; *ΔRF Text Length – CEA*, the change in the number of risk-related words, using the Campbell et al. (2014) dictionary; *ΔRF Text Length – CEA (id)*, the change in the number of idiosyncratic risk words, using the Campbell et al. (2014) dictionary; *ΔRF Text Length – KM*, the change in the number of risk-related words, using the Kravet and Muslu (2013) dictionary; *ΔCosine RF Text*, one minus the cosine similarity of the full text within "Item 1A" between filing year *t-1* and *t*; *ΔCosine RF Text (Heads)*, one minus the cosine similarity of the "Item 1A" risk factors headers text between filing year *t-1* and *t*; and *ΔCosine RF Text (Pars)*, one minus the cosine similarity of the "Item 1A" risk factors paragraphs text between filing year *t-1* and *t*. All word-based measures are divided by 1,000. The cosine similarity design matrix uses raw word frequencies and then applies a TF-IDF adjustment to put less weight on very frequent words. Controls include the full set of control variables used in Table 4. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE 8
Sensitivity Analyses: Alternative Explanations – Controlling for Firm Economics

		Dependent Variable: *VRP*			
	H1	**(1)**	**(2)**	**(3)**	**(4)**
#RF Add/Rem	**(–)**	**–0.24 *****	**–0.24 *****	**–0.24 *****	**–0.25 *****
		(–3.09)	**(–3.16)**	**(–3.13)**	**(–3.28)**
Earn Surprise (t)		0.03		0.03	0.04
		(0.94)		(0.92)	(1.32)
Earn Surprise (t+1)			0.06 **	0.06 **	0.07 **
			(2.19)	(2.22)	(2.40)
Controls		Yes	Yes	Yes	Yes
Match on Surprise		No	No	No	Yes
Fixed Effects		Date	Date	Date	Date
		Industry	Industry	Industry	Industry
N		49,748	49,748	49,748	49,748
Adjusted R^2		0.23	0.23	0.23	0.23

This table presents sensitivity OLS analyses examining the effect of risk factor disclosures on the variance risk premium including additional controls for firm economics. Columns (1), (2) and (3) replicate the main regressions in Table 4 controlling for the contemporaneous earnings surprise, future earnings surprise, and both the contemporaneous and future earnings surprise, respectively. In Column 4 the matched-pairs samples are also matched on the earnings surprise variables. The matching process is described in Table 2. The treatment variable is *#RF Add/Rem*, defined as the absolute number of added and removed risk factors within a 10-K "Item 1A." The coefficients of the treatment variable are multiplied by 100. All variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE 9
Cross-Sectional Analyses: Firm and Risk Factor Characteristics

Panel A: Firm Characteristics

		Dependent Variable: *VRP*					
		Larger Firms	**Smaller Firms**	**High Macro**	**Low Macro**	**Early Filing Firms**	**Late Filing Firms**
	H1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
#RF Add/Rem	**(–)**	**–0.29 *****	**–0.10**	**–0.39 *****	**–0.15**	**–0.36 *****	**–0.11**
		(–3.09)	**(–0.92)**	**(–5.34)**	**(–1.36)**	**(–3.56)**	**(–1.06)**
p-value (1) < (2)		0.02 **					
p-value (3) < (4)				0.02 **			
p-value (5) < (6)						<0.00 ***	
Controls		Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry
N		26,416	23,914	21,772	26,760	23,579	26,750
Adjusted R^2		0.27	0.21	0.24	0.24	0.24	0.23

Panel B: Risk Factor Characteristics

		Dependent Variable: *VRP*					
	H1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
#RF Added	**(–)**	**–0.31 *****					
		(–2.84)					
#RF Removed	**(–)**		**–0.31 ****				
			(–2.42)				
%RF Jump 5+	**(–)**			**–0.08 ****			
				(–2.07)			
%RF Jump 3-4	**(–)**				**–0.05**		
					(–0.83)		
Text Length (RF)	**(–)**					**–0.04 *****	
						(–2.86)	
Text Length (ΔRF)	**(–)**						**–0.01 *****
							(–3.99)
Text Length ("Stale" RF)	**(–)**						**–0.01**
							(–0.77)
#RF Add/Rem Scaled	(–)			–4.36 **	–5.84 ***	–5.49 ***	
				(–1.96)	(–2.69)	(–2.67)	
Controls		Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry
N		53,330	53,330	50,330	50,330	53,330	53,330
Adjusted R^2		0.23	0.23	0.23	0.23	0.23	0.24

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This table presents cross-sectional OLS analyses. Panel A presents results for sample partitions based on firm characteristics. In Columns (1) and (2), we compare subsamples based on firm size, with "Larger Firms" (i.e., at or above the filing-year median market capitalization) versus "Smaller Firms" (i.e., below median). In Columns (3) and (4), we compare subsamples of bellwether firms ("High Macro") versus non-bellwether firms ("Low Macro"), using the technique from Bonsall et al. (2013). Applying this protocol leads to losing approximately 3% of the sample. In Columns (5) and (6), we compare subsamples based on 10-K filing date, with "Early Filing Firms" (i.e., below median days between filing date and fiscal-year end by filing year) versus "Late Filing Firms" (i.e., at or above median days between filing date and fiscal year-end). The treatment variable is *#RF Add/Rem*, defined as the absolute number of added and removed risk factors within a 10-K "Item 1A." The coefficients of the experimental variable are multiplied by 100. Two-sided *p*-values compare coefficients across the partitions.

Panel B reports analyses based on risk factor characteristics. In Column (1) we examine stand-alone additions (i.e., *#RF Added*), and in Column (2) we examine stand-alone deletions (i.e., *#RF Removed*) of risk factors. In Column (3) we examine the incremental effect to the main scaled treatment variable (*#RF Add/Rem Scaled*) of the percentage of risk factors moving by at least 5 positions (*%RF Jump 5+*) in the "Item 1A" of 10-K filings; in Column (4) we assess the same to risk factors moving by 3 or 4 positions (*%RF Jump 3-4*). In Column (5), we assess the impact of text length of the risk factors section (*Text Length (RF)*) and then in Column (6) decompose this text length into that specific to changing risk factors (i.e., risk factors that are added, removed, or that exhibit a material positional change), that we denote *Text Length (ΔRF)*, versus risk factors that do not exhibit significant changes, denoted as *Text Length ("Stale RF")*. The coefficients on the treatment variables are multiplied by 100.

In both panels, *Controls* include the full set of control variables used in Table 4. The matching process is described in Table 2. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE 10
Additional Analyses: Short-Window Tests

	Dependent Variable: *Abn_VRP_Random*			Dependent Variable: *Abn_VRP_Median*		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Three-day [–1,+1] Abnormal VRP						
#RF Add/Rem	**–0.14 *****	**–0.11 ****	**–0.11 ****	**–0.17 *****	**–0.11 ****	**–0.10 ****
	(–3.41)	**(–2.40)**	**(–2.41)**	**(–4.01)**	**(–2.25)**	**(–2.26)**
Controls	No	Yes	Yes	No	Yes	Yes
Earn Surprise (*t*, *t+1*)	No	No	Yes	No	No	Yes
Fixed Effects	Date, Industry	Date, Industry	Date, Industry	Date, Industry	Date, Industry	Date, Industry
N	13,960	13,960	13,960	13,960	13,960	13,960
Adjusted R^2	0.01	0.02	0.02	0.04	0.06	0.06
Panel B: Two-day [0,+1] Abnormal VRP						
#RF Add/Rem	**–0.16 *****	**–0.14 ****	**–0.13 ****	**–0.17 *****	**–0.10 ****	**–0.10 ****
	(–3.04)	**(–2.41)**	**(–2.40)**	**(–3.61)**	**(–2.16)**	**(–2.14)**
Controls	No	Yes	Yes	No	Yes	Yes
Earn Surprise (*t*, *t+1*)	No	No	Yes	No	No	Yes
Fixed Effects	Date, Industry	Date, Industry	Date, Industry	Date, Industry	Date, Industry	Date, Industry
N	13,960	13,960	13,960	13,960	13,960	13,960
Adjusted R^2	0.01	0.02	0.02	0.03	0.05	0.05

This table presents additional analyses examining the effect of added and removed risk factors within "Item 1A" of the 10-K on the short-window abnormal VRP surrounding the 10-K release. The treatment variable is *#RF Add/Rem*, the absolute number of added and removed risk factors within a 10-K "Item 1A." The dependent variable is the abnormal VRP (*Abn_VRP*), the non-event VRP subtracted from the event VRP (following Barth and So, 2014). The non-event window is the three weeks preceding the release of the 10-K, with the non-event VRP using either: (i) a random three-day VRP drawn from the non-event window, denoted *Abn_VRP_Random* in Columns (1) to (3); or (ii) the median three-day VRP across the non-event window, denoted *Abn_VRP_Median* in Columns (4) to (6). The event VRP is the three-day VRP (measured over days [–1,+1] in Panel A) and two-day VRP (measured over days [0,+1] in Panel B) surrounding the 10-K release (the latter being day "0"). *Controls* include the full set of control variables used in Table 4. All other variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on trading dates. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and trading date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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ONLINE APPENDIX

TABLE OA 1
Example of Risk Factor Disclosure "Jump"

Sarepta Therapeutics (CIK: 873303) – Individual Risk Factors Within 10-K "Item 1A"
(Note: selective risk factors chosen to illustrate a jump in positioning)

	2011	2012	2013
(1)	*Our product candidates are at an early stage of development, and it is possible that none of our product candidates will ever become commercial products.*	*Our product candidates are at an early stage of development, and it is possible that none of our product candidates will ever become commercial products.*	*Our product candidates are at an early stage of development, and it is possible that none of our product candidates will ever become commercial products.*
(2)	*If we are unable to obtain or maintain required regulatory approvals, we will not be able to commercialize our product candidates, our ability to generate revenue will be materially impaired and our business will not be successful.*	*If we are unable to obtain or maintain required regulatory approvals, we will not be able to commercialize our product candidates, our ability to generate revenue will be materially impaired and our business will not be successful.*	*If we are unable to obtain or maintain required regulatory approvals, we will not be able to commercialize our product candidates, our ability to generate revenue will be materially impaired and our business will not be successful.*
(3)	*[omitted]*	*[omitted]*	*[omitted]*
(4)	*The Animal Rule is a new and seldom-used approach to seeking approval of a new drug and may not be a viable pathway for seeking approval of our infectious disease product candidates.*	*The Animal Rule is a new and seldom-used approach to seeking approval of a new drug and may not be a viable pathway for seeking approval of our infectious disease product candidates.*	***We currently rely on certain third-party manufacturers and other third parties for production of our drug products and our dependence on these manufacturers may impair the advancement of our research and development programs and the development of our product candidates.***
(10)	***We currently rely on third-party manufacturers and other third parties for production of our drug products and our dependence on these manufacturers may impair the advancement of our research and development programs and the development of our product candidates.***	***We currently rely on third-party manufacturers and other third parties for production of our drug products and our dependence on these manufacturers may impair the advancement of our research and development programs and the development of our product candidates.***	*We have incurred operating losses since our inception and we may not achieve or sustain profitability.*
Risk Factors that "jumped" by 5 or more positions (bolded items above):	**0**	**0**	**1**

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TABLE OA 2
Stepwise Matching Approach and Statistics

	Mean and St. Dev. Differences Between "High" and "Low" Disclosure Samples Based on 5th and 1st Quintiles of Added/Removed Risk Factors					
	Pre-Matching (*N* = 105,068)		**Post-Matching Not Entropy-Balanced (*N* = 50,330)**		**Post-Matching Entropy-Balanced (*N* = 50,330)**	
	(1) Mean	**(2) St. Dev.**	**(3) Mean**	**(4) St. Dev.**	**(5) Mean**	**(6) St. Dev.**
Dependent Variable						
VRP	–0.04 ***	0.03 ***	–0.05 ***	0.01 ***	–0.05 ***	0.01 ***
Experimental Variables						
#RF Add/Rem	14.86 ***	6.33 ***	14.54 ***	6.18 ***	14.54 ***	6.18 ***
#RF Add/Rem Scaled	0.46 ***	0.20 ***	0.45 ***	0.21 ***	0.45 ***	0.21 ***
Matched Control Variables						
MA Ret Volatility	0.03 ***	0.02 ***	0.01 ***	0.01 ***	0.00	0.00
Beta	0.06 ***	0.22 ***	0.04 ***	0.13 ***	0.00	0.00
CFO Volatility	0.01 ***	0.01 **	0.00 *	0.00 *	0.00	0.00
Size	–0.38 ***	–0.05 ***	–0.09 ***	–0.03 ***	0.00	0.00
ROA	–0.07 ***	0.16 ***	0.00 *	0.01 ***	0.00	0.00
Loss	0.19 ***	0.10 ***	0.00	0.00	0.00	0.00
BM	0.08 ***	0.13 ***	0.04 **	0.04 ***	0.00	0.00
Leverage	0.03 ***	0.01 **	0.02 ***	–0.01 ***	0.00	0.00
Analyst Follow	–0.02 ***	0.02 ***	–0.01 ***	0.00 **	0.00	0.00
Inst Investors	–0.01 ***	0.03 ***	0.01 **	0.02 ***	0.00	0.00
Text Length 10K wo RF	0.20 ***	0.02 ***	0.12***	0.04***	0.00	0.00

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This table presents descriptive statistics for matched and unmatched samples of "high" and "low" risk factor disclosers. "High" ("low") risk factor disclosers report a number of added and removed risk factors within Item 1A of a 10-K in the 5th (1st) quintile of the yearly distribution. All variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles. "Dependent" ("Experimental") Variables reflect the primary dependent and independent variables. "Matched Control Variables" include the covariates used to match "high" risk-factor to "low" risk-factor observations. The matching approach is one-to-one "Nearest-Neighbor" (based on "Mahalanobis Distance") and estimated without replacement. Exact matching is performed on year, month, industry membership, and the loss indicator. Columns (1) and (2) report statistics before matching. Columns (3) and (4) show statistics after matching but before entropy-balancing (i.e., observation weights are equal to "1"). Columns (5) and (6) report statistics after matching and entropy-balancing (i.e., observation weights are entropy-balanced with respect to the mean and variance of the covariates' distribution). ***, **, * indicate significance at <0.01, <0.05, <0.10, respectively (two-tailed tests). Continuous variables are winsorized at the 1st and 99th percentiles by filing year.

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TABLE OA 3
Unweighted Full-Panel Regressions

		Dependent Variable: *VRP*				
	H1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
#RF Add/Rem	**(–)**	**–0.27 *****	**–0.22 *****		**–0.09 ***	
		(–3.57)	**(–3.51)**		**(–1.94)**	
#RF Add/Rem Scaled				**–3.59 ****		**–2.14 ****
				(–2.19)		**(–2.29)**
Controls		No	Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Firm	Date Firm
N		180,449	180,449	180,449	180,446	180,446
Adjusted R^2		0.16	0.23	0.23	0.29	0.29

This table presents OLS results examining the effect of risk factor disclosures on the variance risk premium. The sample includes the unweighted full-panel sample observations, reflecting firm-year-month observations with available risk factor disclosures and financial data. The dependent variable is the realized variance risk premium (*VRP*). The treatment variables are *#RF Add/Rem* (*#RF Add/Rem Scaled*), defined as the absolute number of added and removed risk factors within a 10-K "Item 1A" (the number of added and removed risk factors scaled by the total number of lagged risk factors). The coefficients of the experimental variables are multiplied by 100. Controls include the full set of covariates as per Table 4. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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TABLE OA 4
Alternative Specifications

		Dependent Variable:							
		VRP				*ΔVRP*		*VRP (Yearly)*	
	H1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
#RF Add/Rem	**(–)**	**–0.22 *****	**–0.22 *****	**–0.20 *****		**–0.13 ****		**–0.25 *****	
		(–3.56)	**(–3.40)**	**(–3.46)**		**(–2.37)**		**(–3.03)**	
#RF Add/Rem Scaled	**(–)**				**–5.67 *****		**–3.50 ****		**–6.65 *****
					(–3.28)		**(–2.09)**		**(–3.00)**
VRP (m–12)		0.27 ***		0.17 ***	0.17 ***				
		(22.89)		(17.09)	(17.10)				
VRP (pre-filing)			0.33 ***	0.27 ***	0.27 ***				
			(23.34)	(19.60)	(19.63)				
Controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects		Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry	Date Industry
N		48,380	48,380	48,380	48,380	48,380	48,380	4,722	4,722
Adjusted R^2		0.29	0.31	0.33	0.33	0.10	0.10	0.32	0.32

This table presents OLS results examining the effect of risk factor disclosures on the variance risk premium. The sample includes matched-pair firm-year-month (in Columns (1)-(6)) and firm-year (in Columns (7)-(8)) observations, chosen within the highest and lowest quintiles of added and removed risk factors within "Item 1A" (i.e., "Risk Factors") of 10-K reports filed 2006–2019. The dependent variables are: the monthly realized variance risk premium (*VRP*) in Columns (1)-(4), the change between the current month and the prior 12 months realized variance risk premium (Δ*VRP*) in Columns (5)-(6), and the (average) annual realized variance risk premium (*VRP*) in Columns (7)-(8). The treatment variables are *#RF Add/Rem* (*#RF Add/Rem Scaled*), as defined in Appendix A. *VRP (m–12)* is the realized variance risk premium 12 months before the current-month realized VRP. *VRP (pre-filing)* is the realized variance risk premium in the month prior to the 10-K filing date. The coefficients of the experimental variables are multiplied by 100. All other variables are defined in Appendix A. The matching process is described in Table 2. Continuous variables are winsorized at the 1st and 99th percentiles by filing year. All regressions are estimated with an intercept (not reported). "Date" fixed effects are based on year and month. "Industry" fixed effects are based on the Fama-French 48 classification. Cluster-robust-to-heteroskedasticity *t*-statistics are reported in parentheses; standard-errors are clustered by both firm and date. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

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